

TODAY'S SCIENCE FICTION — TOMORROW'S FACT

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STARTLING *stories*

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featuring **TROUBLED STAR** a novel by George O. Smith

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*** START OF THE PROJECT GUTENBERG EBOOK TROUBLED
STAR ***

Troubled Star

A Novel by GEORGE O. SMITH

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FOREWARD—EN SAGA

At least once in every generation there turns up a person who is embarrassing to the Custodians of History. With neither talent nor ambition, nor studious application nor admirable character, this person succeeds where the bright and the studious and the intellectually honest would have failed miserably. Stubborn, egocentric, vain—often stupid—our person blunders in where the wise and the sincere would not dare. His hide is thicker than that of the rhinoceros. He is not abashed to tell the surgeon where to ply his scalpel, or to instruct the statesman on a course of diplomacy. His little knowledge is a dangerous thing—for other people.

His success is due to the law of averages.

History holds many accounts where the brave and the brilliant have stepped in at the right time to avoid disaster. Yet there are more bums than geniuses, more cowards than heroes and more laziness than ambition in our human race, so it is not surprising that there should be occasions when a bum or a self-centered braggart should find that history has a special niche waiting for him.

I

They were parked on the dark side of Mercury, snug and comfortable in their hemisphere of force that kept out the cold and kept in the air. At one side where force met ground, a tall silvery spacecraft rose like a chimney.

They were three:

Chat Honger was tall, red-headed, and thin faced. He looked as though he were incapable of quieting down, but he was really the type of person who has an incredible amount of patience for things which cannot be performed in a hurry.

Bren Fallow was shorter than Chat Honger, darker, stouter, more round of face and more amiable. Definitely, Bren was the methodical type.

The third man was Scyth Radnor. Scyth was the kind of man who is quick to grasp a new idea and as quick to reduce it to practise. His failing was that he seldom looked deep or planned far ahead. Being quick of mind he preferred to play everything by ear because planning required study, and Scyth felt that study for the sake of study consumed too much time—time that could better be spent in the pursuit of fun and games.

Teach them the language and drop them in Greater New York and they would be lost among Manhattan's millions. Better change their clothing, though. Striped shorts, Greek sandals, a Sam Browne belt across a bare chest, and a Roman toga of iridescent changing hues is not the kind of costume seen on Fifth Avenue.

Aside from their costume they were human to the last detail. Even their speech, when translated, sounded like the human tongue. They used slang, elision, swearwords and poor grammar. They made bum jokes and puns. They sounded more like displaced earthmen than technicians from a culture that had been establishing galactic centers of population for thirty thousand years.

"You're certain?" asked Bren.

Scyth nodded. "Dead certain now. It was that last computation that sold me."

"Then I'd better shut down."

Chat Honger shook his head. "We've got a job to do. We're behind schedule now, fellows, because of this question. We've got a beacon to start here, I say let's get along with it and bedamned to the—"

"You can't," said Bren. "The first time you put down in the log that this is a middle sequence flare-star, right smack-dab in the middle of Yalt Gangrow's Diagram, the Bureau of Colonization is going to ask you if you took a look for habitable planets. Then—then what, Scyth?"

Scyth Radnor shrugged. "The answer is 'yes' we took a look and we found one, just at the right distance, the right size, and the right conditioning. To say nothing of upper atmosphere and other data made by observation. So Planet Three is about as habitable as Marandis itself."

Chat grunted. "Looked for any signs of life?"

Scyth nodded. "The phanobands are as dead as you-know-what. The machinus fields are all as dead as one might expect this far from any established route. There are a few bits and dabs of stuff on the radiomagnetic spectrum which show a recurrent pattern too fast to be anything of natural phenomena, however. I say we ought to take a look."

Chat shook his head slowly. "I didn't expect to find it inhabited. But even knowing it is habitable is—"

Bren said, "If mere habitability is all you're after we can go ahead and establish our beacon and leave Planet Three to be handled later. A beacon wouldn't ruin the planet itself, you know."

Scyth said, "We'd better take a look-see anyhow. That last computation on the radiomagnetic stuff looked too much like man-made radiation to me."

Bren Hallow smiled. "Look," he said slowly, "If this planet is inhabited, how come the Bureau of Colonization doesn't know about it. Not one case in the history of Marandis shows the discovery of an inhabited planet that—"

Chat interrupted, sourly, "that didn't stem from Marandian origin. But how about the several cases of spacewreck? Look what we're doing. We're setting up beacons along a rift through the galaxy from Marandis to the Spiral Cluster. We found this rift after years of hard work and galactic surveying and exploring, and both of you know just how fabulous it is. Well, suppose someone found it twenty thousand years ago and got marooned?"

"So what do we do? Take a run to Planet Three and radiate machinus fields all over space? Not until we know. So, Scyth, can you ducky us up a high-sensitivity job out of one of the standard menslators?"

"I think so. D'you think it will work?"

"If there is a primitive culture of the most low-grade organization there, there will also be one or more leading characters. A man of fame or power—or infame and power—whose person will be in the active minds of a large number of hypothetical inhabitants. We should be able to get some sort of response even if the whole thing is primitive as all get-out. But let's take a look before we do anything that's likely to get us into trouble. We're late now, another few hours isn't going to hurt much more."

The discussion in the dome on Mercury's dark side abated as the trio went to work. Scyth began to tinker with his menslators; Chat began to prowl the

confines like a caged animal, thinking deeply, and Bren Hallow went back to his massive equipment that was designed to create a galactic beacon.

On this Third Planet of Sol there were still captains and kings and presidents and commissars and a couple of dictators and a new invention or two, all of which professed to be gentle guardians of the public rights. Only the names had changed, some in violence and some in peace. The names of places were about the same; a few had disappeared in the heat of ideology, but by and large things and people persisted despite atoms, politics and the cussedness of human nature. Youth was still going to hell—and old age was still fuddy-duddy.

One apparent change might have been noticed by a man of the middle of the century, and even he would have expected it.

The history of this change reads like this:

A few years after Global War I, the manufacturer of a breakfast food product known as "Oatflakes" realized a rather monumental increase in the sale of his product. Conscientious investigation showed that this increase was not due to the public becoming addicted to oatmeal as a morning, noon and night diet (with a midnight snack tossed in) but entirely due to a new plaything called the "Wireless." Wireless, it was found, required as a major component about a quarter of a mile of wire wound around the cylindrical box in which the oatflakes were packed.

Some years later, when the first home-manufacture of radio sets slowed because of professional manufacture of commercial radio, the sale of Oatflakes dropped to normal. At this point the manufacturer of the food product realized that the pathway to high sales was not along the contents, but along the package. Let the public buy the stuff for the box, or the box-top. If he wants to eat the stuff on the inside, that's his business!

So in the early-middle years of the century there arose a character called Hopalong Cassidy, who portrayed an Old West chivalry and heroic strength great enough to sell boxtops by the gross ton. He tied-in sales with toy and clothing makers until business reached the Law of Diminishing Returns.

After selling spurs for roller skates the brains ran out of ideas and turned to new fields.

Space travel was the coming thing, so the youth of the land turned to Tom Corbett, Space Cadet.

Tom Corbett's only trouble was the same as the difficulty encountered by one Frank Merriwell fifty years earlier. After twenty years, Tom Corbett became the oldest undergraduate in Space Academy, just as Merriwell became the oldest undergraduate at Yale. The youth of the race wanted a real spaceman, full fledged and heroic, and so they got it.

Meet Dusty Britton of The Space Patrol....

The sleek spacecraft landed and the clouds of hot dust rose almost to the spacelock, driven up by the fierce reaction blast. A hundred yards from the Patrol cruiser lay the broken spacecraft of Roger Fulton, arch-fiend, cornered at last.

The spacelock opened and Dusty Britton looked out through a wisp of the deadly radioactive dust. He was clad in the uniform of The Space Patrol: black breeches and dark blue whipcord shirt piped in gold. Calf-length black polished boots. His head was bare, and the collar of his dress shirt was open wide enough to show the fine muscles of his upper chest and shoulders. He was blondish with a wide open face of the type that is associated with laughing-at-danger. His physique was almost marvelous, slender-waisted, broad-shouldered, long-legged, and agile-armed. His arms and hands and face were tanned from the radiations of Outer Space and there were the million little wrinkles about his eyes that were natural, not because of age, but because of the price one pays for being a Spaceman. At his hip swung the secret sidearm of The Space Patrol, a raygun far more deadly than the Colt .45 in the hands of him who knew its use.

Dusty Britton took a step forward to the edge of the spacelock, took a deep breath, and then jumped down into the floating cloud of radioactive dust kicked up by the landing blast. Within seconds he was out of the cloud again and racing across the ground to the ship of Roger Fulton which had landed askew.

His crew appeared in the spacelock and looked down, not daring to drop into that horror, knowing that they were not as fast as Dusty Britton and could

not make it through in time to be safe.

Across to the wrecked spacer he went, boldly breaching the ruined spacelock. Along the corridor he went warily until he came to the control room. He kicked the door open and walked in, poised lightly on the balls of his feet, lithe and ready to spring like a stalking cat.

Then Dusty Britton faced his arch-enemy, Roger Fulton. Roger Fulton wore a three-day beard, his clothing was stained and torn and his hair unkempt. Fulton watched Britton with cold, angry eyes.

"Now," said Dusty Britton harshly, "Let's have it, Roger!"

Very slowly and very carefully, Roger Fulton's hands found the buckle of his blaster-belt and unfastened it. He let it drop, putting out a leg so that belt and blaster slid easily to the floor. As it reached his toe, Roger Fulton kicked it to one side. He shook his head and sneered at Dusty Britton.

"I should draw and fight the fastest man in The Space Patrol?" sneered Roger Fulton. "I surrender. You'll never blast an unarmed man, Britton!"

Dusty tossed his head. Keeping one eye on Roger Fulton, Dusty sidled across the control room to where Barbara Crandall was tied to a chair. Her eyes were soft for Dusty as he stripped the gag from her mouth and untied her bonds with his left hand. She sat up, rubbing her wrists and working her mouth, trying to tell Dusty something important that would not come through the cramped muscles.

Dusty turned to Roger Fulton. "I've waited for this moment," he said. Quickly he unbuckled his own blaster and tossed it aside. Then he stalked forward, poised to strike, his hands opening and closing at his sides. "Man to man, Fulton. That is, if there's enough man in you to fight!"

Roger Fulton crowed, "Sucker!" and went into whirlwind action. His hand darted inside his shirt and came out with a tiny miniblast.

There came the throbbing sound of raw energy and a flash that blinded. Yellowish smoke curled out and surrounded the scene. Barbara Crandall screamed and tried to get to her feet but the hours of being tied had numbed

her muscles and she fell back into her chair helplessly. The yellowish cloud billowed higher in the control room and began to thin.

Then out of the cloud walked Dusty Britton. He held his right hand by the wrist, shaking it with his left. "Stunned a bit," he smiled bravely.

"But how—?"

Dusty opened the fingers of his right hand and let a miniblast fall to the floor, its charge gone, its usefulness ended. "He tried the old hidden-gun trick," said Dusty. "But two can play that game. Roger Fulton will never menace honest spacemen again!"

The music swelled as the scene faded out; a cheer from Dusty's crew finished off one more opus of Dusty Britton and The Space Patrol.

It was a special occasion, this showing. It was Noon in New Mexico, but the showing had gone out across a worldwide instantaneous network no matter what time it was at the receiving end. In some places it was late in the morning, in some places early, others had this showing late at night. But people were watching back and forth across the face of the Earth.

The film came to end, there was the white flash, then an intermittent flicker as cross-country synchronization took hold. (This flicker was done with an eye toward the dramatic; worldwide networks could latch in without a wink of the screen anywhere in the world.) An announcer came on with the statement that everybody had been waiting for:

"And now we take you to Dusty Britton in person, from White Sands Spaceport in New Mexico!"

A flash and a thundering boom shattered the air and a sonorous voice announced: "X Minus Thirty Minutes!"

White Sands Spaceport was a broad flatland, ringed by thousands of people. In the middle stood a three-stage rocket, waiting; its distance making it look like a small model. In the foreground was a small reviewing stand, and on the stand stood Dusty Britton, resplendent in his Space Patrol uniform. He was extending a hand towards a youngster about twelve, dressed in a

miniature Space Patrol uniform, complete with a miniature edition of the famous "Dusty Britton" blaster at his hip.

The lad saluted Dusty; Dusty saluted back.

Then from his shirt pocket Dusty took a small box and an engraved piece of paper.

"Junior Spaceman Harold Dawson, it is my pleasure to award you this Medal of Spaceman's Honor.

"I am informed that upon July Seventeen, at Thirteen Hundred Hours local time, you, Harold Dawson, Spaceman (Jg) full aware of the dangers that threatened, did without thought of your personal safety, wade deep into the shifting sands of Mudlark Lake and from that deadly quicksand return your smaller sister to safety. For valor and for gallantry, I present you with the Order of The Golden Heart!"

With a flourish, Dusty pinned the decoration on the proud youngster's chest. The medal glittered there, a small heart of gold surrounded by rings like those of Saturn, carved in flat relief.

Then with another exchange of salutes, Dusty Britton went down the steps and into a waiting spaceport jeep and while the crowd cheered wildly, Dusty was driven across the sands to the spacecraft.

With tolerant parents permitting their young to watch this live, in-person show no matter what time it was across the earth, it is not hard to believe that during these many minutes there were more people thinking about Dusty Britton than there had ever been people thinking about any other person at any one time in the course of history.

And so Scyth Radnor, tinkering with his menslator on Mercury, trying to tune it to some response that would deliver definitive thought, caught much more than he anticipated. In fact, it nearly overloaded the device.

"Any doubt?" he asked with a twisted smile.

"Nope," from Bren.

"I pass," added Chat.

Scyth said, "So instead of being an uninhabited planet, we have a rather high culture, complete with space travel. This Dusty Britton must be quite a hero.

But how in the name of the Great Space can they have space travel without machinus fields or some knowledge of phanoband radiation?"

"Maybe their space travel is—er—"

"Now look, you're not suggesting that people with a Space Patrol are riding ships with tailburners? Rockets? What a horrible thought."

Bren shook his head. "Our forefathers lived through it."

"Not many of them," grunted Scyth.

Chat objected. "Read that history you dislike so much. You'll find that our ancestors went through hundreds of years wallowing across space to the planets in reaction-type spacecraft. Chemico-atomic rockets, if you please."

"Let's stop the argument and get along with the main problem," said Bren. "What are we going to do about them?"

"Well, we can't set up a beacon with them here. So we'll just have to take the proper measures."

"That'll be quite a project. Whole colonies and—"

"That they haven't got yet. They're at the outpost stage; the scientific expedition stage. Their moon has less than a hundred people on it, their Mars has been visited only three times, and their Venus only once previously. This project that Dusty Britton is going on is the second Venus rocket, the first one being sent as an orbital, round-trip manned-job for observational purposes. So we can set up our barytrine field without causing a lot of distress, and then we can go on preparing our space beacon."

Bren nodded and Chat said, "You're the handiest man with menslators and the like, Scyth. You're also the guy that can think fast on his feet. We elect you to go to the Earth and contact this Dusty Britton and explain to him so that he can tell his people what is going on."

Bren nodded. "Take the ship and go, Scyth. But use the driver as little as possible. We'd still like to keep this rift secret, you know. We're working for Transgalactic, not the whole damned shipping business."

Not long after, on its secondary drivers which did not radiate enough to make direction-finding much better than haphazard, the spacecraft rose from Mercury and headed toward Earth.

II

Dusty Britton entered the lower cabin of the three-stage rocket and flopped into a chair. "Quite a show," he said with a trace of scorn.

Martin Gramer, the producer of the long series of Dusty Britton pictures puffed his cigar and nodded with self-satisfaction. "Not bad," he said. "Not bad at all."

"Gramer, how the hell long is this nonsense going to go on?"

"Until you're ready to retire."

"I'm ready now."

"For good?"

"I could do something else, you know. After all, I am an—"

Martin Gramer eyed the husky young man with derision. "You say 'actor' and I'll blow a gasket," said Gramer.

"Then what the hell am I doing here?" roared Dusty.

"You're here because you have an honest-looking face and a pair of broad shoulders to go with it. You're the living embodiment of John Darling Trueheart, and you can act the part, providing some bright guy lays out the floor plan and coaches you."

Dusty growled, "Why not hire the bright guy?"

"Because he's got a face that would scare children and the physique of an underfed fieldmouse. Pull you out of that hero role you're in and you'd fall so flat on your face that folks would be calling you Old Doormat. Now snap out of it, Dusty, and be glad you've got hold of a good thing. Stop looking for something you couldn't handle."

Angrily Dusty got up out of his chair. "I suppose you think it's fun to have to go roaming around the country wearing this jazzed-up surveyor's suit with a three-pound chunk of rusty iron clanking on my hip."

"To date they've sold three and a quarter million replicas of that Dusty Britton Blaster you're so contemptuous of, and you've received ten cents for

every one that crossed the counter. What's so damned bad about that?"

"I feel silly."

Gramer roared with laughter, then cut it to one short bark as he cooled down to eye Britton angrily. "What's so damned silly about being a model of honor and respect for several million kids?" he demanded.

"Did you ever think how imbecilic it sounds to be Dusty Britton of The Space Patrol, with no space to patrol, wearing a blaster that doesn't blast? And wearing a pack of medals stamped out in the model shop? What does it all add up to?"

Martin Gramer tossed the stump of his cigar at the disposal chute and faced Dusty with a hard expression. "It adds up to a lot, Dusty. It adds up to a damned good living for you. It adds up to—maybe something you're too dumb to understand, but I'll spiel it off anyway—being an ideal. Damn it, man, there's millions of kids in this world that eat, think and dream about the Space Patrol and Dusty Britton. You're an idol as well as an ideal, Dusty. Kids follow a big name man. It's a darned sight better that they follow an ideal rooted in virtue, strength, honesty and chivalry than to have them trying to emulate characters like Shotgun Hal Machin or Joseph Oregon."

"Yeah," drawled Dusty, "But do you know what it means?"

"You tell me your version, Dusty. As if I hadn't heard your gripe before."

The disgruntled actor took a deep breath, opened his mouth, but then closed it again. He let out most of the blast he was preparing and said, quietly but disgustedly, "Why waste my breath? Dusty Britton doesn't smoke. Dusty Britton drinks soda pop and milk. The only women in Dusty Britton's life are his aged mother and his younger sister. Dusty Britton's biggest gamble is when he offers to bet a Saturnstone on this or that. Hell's Eternal Fire, Gramer, do you realize that I can't even date a dame for a dance because 'Kids don't care for the mush stuff!' and my private life is not my own? I can't even swear, god-dammit!"

Gramer eyed Dusty cynically. "You seem to get along."

"Sure. I get along. When I shuck this monkey suit and dress like a human being. But you know what happens? When I turn up at some joint, do I get introduced as *The Dusty Britton*? Like hell I do. I'm treated like any of the rest of the dopey tourists. Herded like cattle to the rear seats, while a tomato like Gloria Bayle luses in with her fourth husband and gets the works on the house."

"You make my heart bleed, Dusty."

"Your heart never bled anything but vouchers," snapped Dusty. He fumbled in his hip pocket and pulled out a flask.

Gramer did not say a word.

"Well, aren't you going to give me an argument?" demanded Dusty.

"No. You can't be seen."

"But someone's likely to smell bourbon on my breath."

"No one that counts. And by the time we get back—"

Dusty stopped raising the flask in midair. "Get back—?" he roared. "Get back. Look, Gramer—"

"Sit down, Dusty. Take it easy."

"Gramer, what goes on here? You're not suggesting that we take off in this fire-breathing hot water boiler, are you?"

"You've read all the advertisements."

"Yeah, but nobody with sense would take ad-writer's copy for anything but guff."

Outside, a bomb burst with an ear-splitting racket. A stentorian voice thundered, "X Minus Five Minutes!"

"Ye Gods, you're really going through with this madman's publicity scheme?"

Gramer smiled. "Sure. It's just to Venus; but you can bet your life that every kid that sees this take-off on video or here on the field will be dreaming of the fabulous adventures you'll be having. Those kids *know* this is for real, Dusty."

"Include me elsewhere," mumbled Dusty. He started for the spacelock.

"You can't let those kids down!" roared Gramer.

Dusty paused at the sill of the spacelock. "Gramer," he said cynically, "I'm not letting anybody down. I'm just keeping the hide of Dusty Britton in one unscarred piece."

"But the public—"

"That's what you've got press agents for, Gramer. So you can get your high-priced publicity men to run a few miles of paper explaining how I happen to have left this shooting star four minutes before take-off!"

"Dusty, you're a no-good louse."

"But a whole one. And let me tell you this, Gramer, you're less worried about the state of youthful morals than you are about losing the thread of a good, high-selling series. So I'm going to sail out of here as though I was scared to death of rockets—which I sure as hell am—and you're going to tell some bright explainist to get busy earning the dough you pay him. And when the smoke is all cleared away, I'll be safe and you'll be safe, and Dusty Britton will continue to go rolling along and the box office will continue to come rolling in. Spend a few short months in space? Not while the geegees are running at Hialeah!"

"But Dusty—"

"Space? Bah! Nothing, floating gently from vacuum to void and back again. Not for Dusty Britton!"

Dusty paused long enough to run splayed fingers through his hair and then he headed for the spacelock with a determined step.

"Wait!" roared Gramer.

Dusty paused.

"The least you could do is to go out of here not looking like Dusty Britton. Don't be an ass! I'll cover for you, but you've got to help!"

"All right but—" Outside another bomb racketed and the amplifier announced laconically, "X Minus Three Minutes!" and startled Dusty with the realization that he did not have much time. "—make it quick!"

"You—there!"

A technician coming up the ladder looked startled.

"Fifty bucks to swap clothing with Britton, here."

"Done," and the tech started to peel. He balked at Dusty's famous 'Blaster'?

"That's worth another—"

"Another fifty—dammit!" agreed Gramer. "Now, wave out the door while Dusty leaves."

The roar that went up was for their beloved hero waving out of the spacelock, not the tech that came down the ramp with a rush, followed by the portly Martin Gramer. The spacelock swung closed as the spaceport jeep pulled away with Dusty and Gramer in the back.

They were a half mile away when the thunder came. No one even noticed them wending their way through the crowd, for every eye on the field was looking upwards, straining to see the spacecraft that was carrying Dusty Britton and The Space Patrol off to new adventures.

About a hundred miles off the coast of Baja California, Scyth Radnor sat in the control room of the big spacecraft. The dome was awash. Scyth sat high in the dome watching the pleasantly lazy progress of a forty foot schooner that was coming in his direction. It was a pretty sight and Scyth appreciated it even though he had been born on Marandis some thirty thousand years after the sail as a functional device had been outmoded. Sail, to Scyth, was strictly a vacation sort of thing, just as it was to Dusty Britton and a few billion other people whose lives are geared to a time-table except for vacation time.

If there was any puzzlement over this, it was because Scyth's menslator was not following the rocket, now laboring in free flight towards Venus. Dusty, according to what Scyth had been able to pick up, should have been there instead of here. But Scyth was not the burning inquisitive type. He knew that there was some explanation and that he could afford to wait until it was given instead of wasting a lot of energy trying to figure out the motives of a member of a race unknown to him.

He had better things to contemplate.

In the field of his telescope he could see a sight he approved of.

It was not Dusty Britton, lazing easily near the wheel of the schooner, keeping the helm steady with his left foot because his hands were occupied with a drink on one and a cigarette in the other. It was Barbara Crandall, lying on the cabin on a blanket. Her ankles were crossed and the arch of the upper foot was high and graceful. One thigh, slightly higher than the other, glinted from the sunshine, dark tan. Her breasts pointed at the sky, molded in dazzling white that contrasted sharply against the healthy, animal tan of her flat tummy. There were many more square feet of healthy hide showing than there were of the white shark-skin affair she wore, and Scyth approved of the view.

As he watched her, Dusty drained his drink, tossed his cigarette overboard, and called:

"Hey, Barb! Get us another quart, will you?"

Scyth did not hear it, for his menslator was by no means that competent a device. He just watched and wondered what they were saying.

Barbara called back, "Out of it already?"

"Yeah. I'd get it myself but someone's got to drive this rig."

"Don't mind." She stretched languorously and stood up, stretching high; pulling in her stomach and arching her back with her arms stretched high above her head. Scyth whistled inadvertently as her body went taut against the wisps of dazzling white that crossed her breasts and hips. She came along the cabin top, dropped into the cockpit, and disappeared into the cabin. She came out a moment later with a bottle which she opened and handed to Dusty. She took the wheel while he poured. They toasted one another. They sat side by side, their shoulders touching.

"Nice," she said quietly.

"You bet."

"Nice, quiet and peaceful."

Dusty addressed his glass and held it high. "Here's to the G. D. Space Patrol."

"What are you supposed to be doing?"

Dusty laughed. "I don't know. I'll find out when we get back. Gramer will have some flanged-up explanation right and ready for me."

"You'd better hope that the G. D. Space Patrol doesn't catch you all at sea with me."

"Phooey," he said. He pursed his lips and Barbara gave him a gentle peck that made Scyth's blood bubble slightly.

"Phooey nothing," she said. "You'd be—er—cashiered. Imagine a member of The Space Patrol consorting with a woman."

"What's good enough for pappy is good enough for me."

Barbara chuckled knowingly. "Where are we heading, if it's of any importance?"

"There's an island dead ahead. We might camp on the beach for the night. It's fine clean sand and—"

"You mean that hummock over there?"

"Hummock—hum—Good Lord!"

The hummock, dome of Scyth's spacecraft, began to rise out of the sea. Yard after yard it rose, coming upward glistening wet, the sea water running down in rivulets along its sleek flank. Ponderously and inexorably it rose with a steadiness of living rock. Yet it carried the air of feather-lightness, of an untold monster of sheer power held in easy leash. This was no rocket, straining against the formidable pull of gravity; this was a thing above material forces, its engines idling, its control in complete command. Without a second glimpse it was no spacecraft of Earth.

Up out of the sea it rose until its hundred yards towered above them. The spacelock was just above the waterline when the rising stopped and the alien spacecraft stopped, rock-steady. It was poised on its inexplicable driving forces with the same confident ease that an elevator shows when poised on

its cables at the twentieth floor of a building. It stood rock-still and let the ocean waves break against its sleek, polished metal flank.

Whatever it was, Dusty did not like it.

He kicked the auxiliary engine into life, loosed the halyards and let the sails drop. He turned the helm hard as the engine roared into full throat. But the schooner defied its helm and aimed bowsprit-on to the spacelock of the spacecraft, starting through the sea like a dolphin toward the ship of space. The engine raced without bite because the ship was being hauled forward by some unknown force faster than the screw could drive it; the helm shuddered but had no effect, it tried to slue the stern sidewise but only succeeded in making the hull strain out of line. The wheel whipped out of Dusty's hand and spun to dead-ahead.

Dusty left the helm and dived into the cabin. He flipped on his radio and waited with rising panic while the tubes warmed and the meter rose to the red line that meant that it was ripe and ready for use. He grabbed the microphone, flipped the bandswitch to the Coast Guard Frequency, and yelled:

"This is Dusty Britton of the schooner Buccaneer. We are about a hundred miles off the coast of Baja California. Help! We are attacked by an alien spacecraft! Help! This is—"

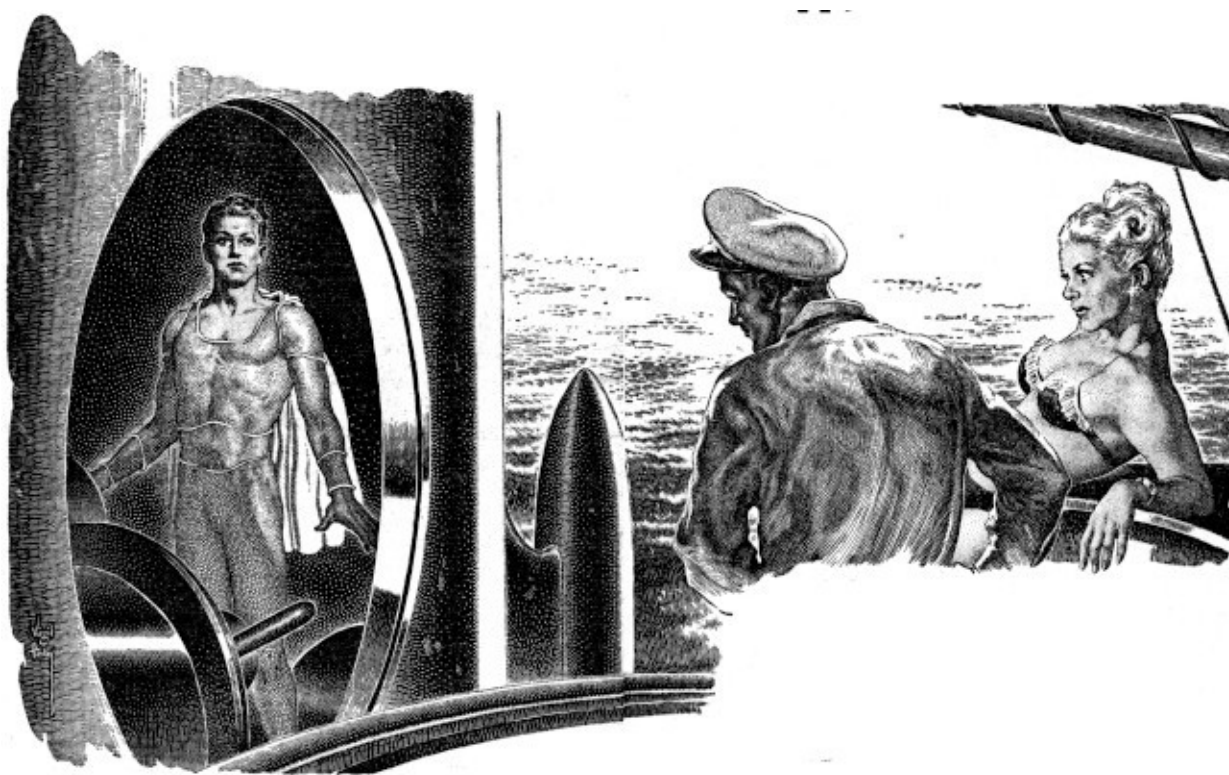
He let his voice trail off because the output meter dropped abruptly to zero. Something had gone kaput.

III

Dumbly frightened at the face of the unknown, Dusty was far more frightened at being confined in the cabin of his schooner than he was of the nameless horror he would have to face above. He left the cabin in a hurry, and with mental desperation he turned deliberately to face the danger in the hope of getting it over with. He figured there would be less anguish if it came quickly.

The spacelock door was open wide and a man was standing there with a fluted-barrelled thing in his hand. On the deck were droplets of copper still

hot enough to send up little wisps of smoke from the deck. The stub end of the antenna was melted down in a blob. As Dusty looked from Scyth Radnor to his ruined antenna and back again, Scyth leaned back in the spacelock and dropped his weapon. Then he made a relaxed show of sitting on the sill of the airlock with his feet dangling almost to the tips of the waves. He looked relaxed and calm and the trace of a smile was on his face; the kind of smile that would open into honest pleasure if he were greeted with the same.



"I am sorry," he said. "I am Scyth Radnor of Marandis. Despite the fact that I was forced to ruin your antenna, I do come on a peaceful mission, Dusty Britton."

"Yeah—" mumbled Dusty stupidly. Barbara was leaning flat against the mast, white-faced under her tan.

"Believe me, Dusty. I mean no harm. I did have to prevent you from broadcasting that which would bring a bad impression of me to your people."

Scyth reached up and pressed a button in the wall of the spacelock above his head. The sill of the spacelock came out abruptly in an extensible runway, carrying Scyth forward over the deck of the Buccaneer. Scyth dropped to the deck and stood facing Dusty with a hand extended.

"What do you want?" stammered Dusty. "And how come you talk our language?"

Scyth pointed to the tiny case slung around his neck. "This is a menslator," he explained. "When used in direct conversation with a man of another tongue, it acts to translate for both parties their meaning. It isn't perfect by any means, but it does help to make people of different tongues understand one another." Scyth smiled and then said, "For a quick and amusing explanation, observe this." Scyth clicked the switch off and began to speak. His speech was utterly comprehensible to Dusty and Barbara at first, but Scyth clicked the little switch after he had said a few words. They heard Scyth like this:

"Fa d snall id, an expression meaning to consign to the region of theological punishment, which when repeated through the menslator becomes 'Go to hell!' See?"

Dusty nodded dumbly. Barbara relaxed slightly.

"Now," said Scyth, "I am from Marandis. Marandis is a planet only a few thousand light-years from the Galactic Center, which makes it nearly thirty thousand light-years from here. Marandis is the seat of the Galactic Government. Look, Dusty, I came here to explain all this to you. There is a lot to say, and there is a lot you must take on faith until you know all of it. Let's relax. Will you come aboard my ship and have a drink? It's comfortable there and—"

"No!" snapped Dusty.

"Why not?"

"Nobody, but nobody, is going to get me in any space ship," said Dusty positively.

Scyth eyed Dusty queerly. His thoughts would have been obvious to anybody but Dusty and Barbara. Scyth was trying to justify in his own mind the attitude of a High Brass in The Space Patrol (*any* space patrol) who would not enter a spacecraft. Scyth finally decided that Dusty's reticence was due to Dusty's suspicious nature. Dusty was unarmed and he was not getting into a spacecraft capable of carrying him across the galaxy, perhaps operated by other members of the crew. There were no other members, but the ship was big enough to have many. Scyth nodded to himself and smiled at Dusty.

"As you prefer. I only repeat that I mean no harm and I add that the salon inside is pleasant. We can all have a—"

"We've got a drink," blurted Dusty. He turned on his heel and got the quart from the seat by the helm. He stopped to get a third glass. He poured.

Scyth tasted gingerly. "Very smooth," he said. "What is it?"

"Bourbon."

"Bourbon. Tastes like an excellent liquor. Thank you. Now—" Scyth sat down on the edge of the deck with his feet hanging into the cockpit and settled himself for a session. "Dusty, we are here because we are creating a beacon for our galactic spacelanes."

"Beacon?"

Scyth nodded. "You have the insular viewpoint," he remarked. "You can stand at night and point out your destination. But you cannot even see Marandis from here, even with the finest telescope ever built. Stars lie in the way, huge gas fields and nebular clouds block fast direct passage. To chart our course safely past such stellar menaces, we establish beacons at the ends of certain free passages. For instance, Sol lies at the end of a fifteen hundred light year straightaway from the last beacon we set up. Here at Sol a slight turn in the course is made and there is another straightaway for a thousand light-years toward the Spiral Cluster. We—my friends and I—are charting the course through a rather interesting rift from Marandis to the Spiral Cluster. This rift, along which you lie, has been hidden from us for

thousands of years. When it is finished it will cut hours from our travel-time."

"And maybe so. But what is a beacon and how do you establish it?"

"Dusty, when a spacecraft is running at fifteen hundred light-years per hour, a three-day-variable star winks in the sky ahead like a blinker-light." Scyth chopped his left palm rapidly with the edge of his right hand. "Wink-wink-wink it goes. And the pilot puts his spacecraft point-of-drive on the beacon and holds it there until he passes it and aims to the next. You—"

"Variable star!" blurted Dusty.

"Yes. The three-day variables are used for course markers; the longer variables are used to denote gas fields, nebular dust, and the like, and the still-longer beacons are used to denote places where various well-travelled starlanes meet, cross or merge. It is—"

"Three day variable—" breathed Dusty.

"Yes. In three days Sol will rise ten times its present brightness and fall again to less than one tenth of the present brightness. This is accomplished by creating an atomic instab—"

"My God! How can any race live under such conditions?"

"They cannot. Not unless properly prepared, well taken care of, aware and ready for it."

"Look," snapped Dusty. "Why not go out and use some other star for your damned beacon?"

Scyth shook his head. "If we were gods," he said quietly, "we could park the Galaxy on our desk, pick up a broom-straw and by fitting and trying we could locate the best course through the star-fields. But—"

"If you were gods," grunted Dusty bitterly, "you could reach in and move a few stars aside and run your damned channel on a dead line from one end to the other. So why do you use Sol?"

"Because the two straightaway lanes that meet at Sol do not meet at some other star. In one or two cases along this rift the original surveyors provided alternates in case we ran into trouble. But not on this one. No, Dusty, we cannot change our plans."

"But see here—"

"Dusty, you wouldn't stand in the way of Galactic Civilization, would you?"

"You're damn well tootin' I would if it's going to mow me down if I don't."

Scyth said soothingly, "Doubtless you have cases on your Earth where a state highway is surveyed right through someone's home. Tell me, Dusty, what happens then?"

"We buy the property at a fair price so that the family can find another home of the same value."

"So you don't stand like a barrier in the way of advancement."

"No we don't. But where are we—" Dusty eyed Scyth with a frown. "You're not going to tell me that your gang will migrate the people of Earth to another solar system, lock, stock and barrel?"

"That would be impossible, of course."

Dusty grunted. "So we gotta alternately cook and freeze just so your outfit can run a goddamned traffic pattern through our living room?"

"Well, now, it's not that bad," said Scyth placatingly.

Dusty did not hear the Marandanian. He was thinking of Los Angeles suffering under the effects of a variable star. Or, rather, he was trying to visualize such a condition. His imagination provided alternating scenes of icy blast and deadly heat, but Dusty's overall technical knowledge was far too meager to offer him even a slight glimpse of the real truth. To merely consider Sol varying about one hundred to one in brightness and warmth every three days was as far as Dusty could go. What would happen to the weather, the general climate, agriculture, and all of the rest were far beyond Dusty.

Even so, the sketchy picture provided Dusty with enough data to say, "Why, we couldn't go on living on Earth at all!"

"Right. Which is why I'm here."

"But you said—"

Scyth smiled confidently. "I'm not here to preside over the death of your part of our human race," he said. "I—"

"Our part of your human race—?" exploded Dusty.

"Of course," said Scyth in a matter-of-fact tone. "So far as we know, human life was first spawned on Marandis. About thirty thousand years ago we became galactic in scope, spreading out, colonizing, expanding, exploring. Many expeditions left home and were lost. But I'll not belabor this any more, just accept my word for the following: nowhere in this galaxy have we found intelligent life that did not spring as an offshoot of misplaced Marandian culture."

"How can you be so damned certain?"

"The easiest way is to check the cross fertility. It has always worked, to date at least," said Scyth, inadvertently letting his eyes slide up and down the very pleasant sight of Barbara Crandall's body. Barbara knew Scyth's contemplative look and she reacted as any uninhibited woman does when some man is measuring her. The deep high breath raised her breasts and flattened her stomach even though she had no great yen toward wanton promiscuity.

"I gather, then, that you and your gang are going to do something about us?" she asked.

"Of course. We have a program for cases like this. Since you cannot live on a planet rotating about a variable star, we'll move Earth to another star of the same classification."

"But—" objected Dusty.

Scyth went on as though he had not been interrupted. "We'll set up a barytrine field around Earth which serves to do two things. A barytrine field cuts the force of gravity that holds Earth to Sol. It also produces a complete stoppage of objective and subjective time within the field. Then with machinus force-fields we'll put Earth in motion towards another star of Sol's general size. In a thousand years you'll come out of the barytrine field and resume your daily lives under the light of a brand-new sun. It's as simple as that."

Dusty eyed Scyth sourly. "Maybe I've got this wrong," he said. "Maybe you think we live a hell of a lot longer than we do. Maybe you live a thousand

years and more but we—"

Scyth held up a hand. It was the hand that held the glass, which was empty. Dusty, reacting as he always did to the sight of an empty glass, filled it despite the fact that he felt that Scyth Radnor was a long way from being a friend.

The visitor from space smiled indulgently. "You miss the point, Dusty," said Scyth, nodding his thanks for the drink. "I said that the barytrine field produces a complete stasis in time. It will snap on ... a thousand years will pass ... it will snap off. To us, we will live and die and never see you again. But for you and yours, if you drop a marble before the field goes on, time will cease for you until the field goes off, and your marble will hit the floor a thousand years from now. You will feel nothing. There will be a tiny flick of light. If you are watching the sun it will probably blink and return slightly off-center because we never can be that precise. If you are watching the stars at night, they will wink out and wink on, and be in a new pattern. You will feel nothing."

"Yes, but, look here, we—"

Scyth smiled again. "Oh, you'll be repaid. We'll raise you from your present primitive level—"

"Primitive?"

Scyth nodded. "Primitive," he said. "You're as primitive to us as your savages are to you."

"But—"

"Look, Dusty, thirty thousand years ago, Marandis was still ahead of your present state of development. I can say this because your people at the present time still have no inkling as to the inconsistencies in the theory of general relativity. Someday soon you will discover that general relativity does not fit all the cases. Then you will propose the machinus theory of space-time. The machinus theory works where relativity does not. Then," glowed Scyth, "you will discover the phanoband carriers which operate in a

way as to completely deny relativity in every concept. From there you find the barytrine field forces. But you're still primitive, Dusty."

Dusty eyed the Marandanian sourly.

Scyth continued, "You'd find little in common with us," he said. "You'd find that you would have to re-educate yourself before you could even understand us. Why, there are people in our culture who would take advantage of your ignorance."

Dusty nodded. His hazy knowledge of history presented him with a costume drama of Sir Walter Raleigh handing over a ten, two fives, and four ones to Chief Sitting Bull and receiving in return an engraved bill of sale for the Island of Manhattan. This negotiation was sealed with a slug of liquor out of a bottle labeled 'Bourbon, Bottled in Kentucky.' (Pocahontas, standing to one side, received a string of beads.)

Scyth went on:

"The big problem, Dusty, so far as you are concerned is the preparation of your people. We cannot be precise about the position of the new sun. We could not possibly hope to keep any semblance of your stellar geography. When the barytrine field goes on, it will produce an effect similar to reaching the splice in a reel of film. With no warning, no pain, strain, nor furor the sun will snap slightly aside to its new position. On the night-side the stars will flick instantly to a new pattern. This sort of change would cause great hysteria and fear. Unless the people are prepared for the sudden change. So, Dusty, you as a high official in your Space Patrol must carry our message to your people."

Dusty said, "But—"

"You've mentioned the possibility of payment," said Scyth smoothly. "We expect and intend to pay. But not in money, Dusty. In service and commerce and in many other ways. For instance, we know that your group—I cannot call it your 'race' because your race is ours—must stem from an early expedition and so you are a lost offshoot. As soon as we can, we will come to you with teachers and learned men to help you regain your rightful place as a part of our Galactic Culture."

Dusty looked at Scyth. In his mind churned a hundred objections to the whole thing. He did not like it at all, but he was logical enough to realize that

his objections would be waved aside and the Marandanians would go on and do as they planned anyway. On the other hand, maybe if Dusty Britton were to take a large hand in this affair and carry it off successfully, Dusty Britton could become a large figure indeed.

"It will be a bit difficult," he said slowly. "People are not going to take to the idea of losing their sky and sun and a thousand years out of the middle of their lives."

"The thousand years are peanuts. Nobody will notice it. The swap in suns is only a sentimental objection. One sun is like the next and we'll see to it that they are as close as can be had. The change in stellar appearance is deplorable, I admit. But it will give you one advantage, Dusty. Like most skies, they are divided off into primitive legendary shapes with neither rhyme nor reason. A cluttered mess. With a fresh start you can make some reason to the constellations. These are the sort of arguments you must use, Dusty. As a final reminder, you must remember that this is what is going to be done. Period. It is necessary and it cannot be stopped. Therefore you and your people should accept it and make the best of it. Therefore, in what will seem like three weeks, you will be by another star, under a strange sky, a thousand years from this moment. And my people will be there waiting to help you on your climb to the pinnacle of culture.

"But now I must go. Take my words back to your leaders, Dusty. You will go down in history; make the best of it!"

As abruptly as that—Scyth Radnor arose from the deck of the Buccaneer, climbed onto his runway, and was drawn back into the big spacecraft. The spacelock closed smoothly and the huge ship rose silently out of the sea and arrowed towards the high blue sky. The only noise was the whistle of its passage through the air above.

Scyth landed beside the bubble on Mercury's dark side not long after. Chat greeted him with a question about his success and Scyth smiled. "Naturally they didn't cotton to it," he said. "No one ever would."

Chat nodded agreement. "They wouldn't stand in the path of advancement, would they?"

Scyth chuckled. "I'm getting to be something of a diplomat," he said. "Not good, but I think adequate."

"Yes?"

"Sure. First I told them about the beacon and let them ask questions about it to whet their curiosity. Then I explained what the beacon was, which horrified them completely, as it should. Then after letting them cook in their own fright for some time I let them down easy by explaining how we would help to save them. So now there's nothing to do but to finish off the job."

"Right. How long will it take for you to get the barytrine generator set up and ticking?"

"Call it a couple of weeks. I'll have to go back to Marandis for the generator. It may take me a day or two to get it, you know. We'll have to get our license revised, and we'll have to put a bond against the safety of this planet Earth, as they call it. Of course, we'll have lots of time to look for another sun where we can put their planet; we can do that after the beacon is started and they're out of danger-distance."

Bren said, "So the first thing for you to do is to hike back to Marandis and get your barytrine generator."

Chat added, "When you take off from here, be sure you go due North until you're a long way out of line. No use in advertising our position."

"Right. I'll fog-off the course as best I can."

IV

Within a few minutes after his return to Mercury, Scyth Radnor was on his way back to Marandis to make the final arrangements. He took the long way out of this part of the galaxy and wound his way in an inextricable pattern to confuse any possible competition. Until the through-route was surveyed and the first passage made from end to end, there would be no exclusive franchise; another company might be able to latch onto one open lane on this route and give them competition.

Considered as unimportant was the fact that Scyth Radnor took along with him the beefed-up menslator that had put him on the mental trail of Dusty Britton. Not that this mattered, the chances were almost perfect that no one of them would have done anything with it anyway now that their problem was settled. At least, not Chat or Bren. Scyth might have played with it in an off moment. He alone had gotten an eyeful of Barbara Crandall, and while Barbara seemed to be Dusty Britton's woman, Scyth might have wondered whether there were any more at home like her.

But Scyth was on his way to the galactic center, out of range of menslators, even the big permanent installations.

Scyth, Chat, and Bren are not to be criticized for leaving a job undone. To them, a mere explanation covered the entire program. They did not expect the natives to understand the complex ramifications of the galactic culture any more than a certain native chief could understand the danger of fishing in Bikini Lagoon some fifty years earlier.

In fact, the three of them might have been highly amused at a primitive culture that had committed the egregious error of placing such a high value on something of no intrinsic value.

But back on Earth, the wires buzzed and the headlines screamed, and a brace of Gramer's press agents were hard put to untangle the mess the Marandanians had started.

From the teletypes of Worldwide Press Service:

UNITED STATES COAST GUARD RADIO TODAY
REPORTED A DISTRESS SIGNAL FROM SCHOONER
BUCCANEER OFF COAST OF BAJA CALIFORNIA STOP
BUCCANEER ATTACKED BY QUOTE ALIEN
SPACECRAFT ENDQUOTE STOP USE WITH DISCRETION
COMMA BUCCANEER OWNED BY DUSTY BRITTON OF
MARTIN GRAMER STUDIOS STOP

An excerpt from the daily column of Garry Granger:

"There is something in the wind that smells like a publicity stunt. Dusty Britton, our Space Patrol type Sir Galahad supposedly took off for the Venus jaunt some three weeks ago, but has succeeded in sending a distress signal from somewhere off the coast of Southern California. Apparently The Space Patrol is about to meet up with Moby Dick, or possibly it will be "Ten Thousand Leagues Under The Sea" starring Dusty Britton. We would like to know two things: one is whether our intrepid hero actually risked his million dollar neck in a rocket or not, and the second thing is how much hanky-panky the Coast Guard is going to stand for. Some things should be kept sacred. We are not very religious here at the office; but we do believe in the Brotherhood of Man, and somehow we resent bitterly the use of distress signals as a means of getting publicity."

Excerpt from a press release from Martin Gramer Productions, Inc.:

"Now it can be admitted! Dusty Britton has combined fact with fantasy! No longer a mere actor, Dusty Britton was called from the space rocket just a few minutes before take-off time to investigate a secret report of space operations off the coast of Baja California. If Dusty Britton reported an attack, it stands to reason that the secrecy that surrounded the original report is no longer necessary and Dusty Britton's presence on earth instead of in the space rocket can be disclosed. We await more detailed information as to the real nature of—"

From a press-conference held at Arlington, Virginia:

SIGNAL FALSE! SAYS F.C.C.!

"Radar Stations report that no sign of space operations by any agency other than the Venus Rocket have been observed. Even the early warning screen operating along the coast of California and Lower California has nothing to report. The signal of distress is obviously false, and Dusty Britton will be asked to show just cause for emitting such a report."

A statement from the United States Coast Guard:

"Search and rescue squadrons of the Coast Guard were in flight above the schooner *Buccaneer* within an hour after the interrupted distress signal from *Dusty Britton*. The schooner appeared to be in excellent condition and was making its way back towards land when sighted. Radio challenges were ignored but upon flying low, *Dusty Britton* and an unknown woman were seen waving from the deck. There seemed to be no signs of distress, but a Coast Guard cutter is speeding to the ship and is expected to make contact in the next few hours."

Excerpt from the column of Garry Granger:

"What actor, long noted for his derring-do and his exemplary behaviour has been in unchaperoned company with a nubile young female in romantic surroundings? In our youth, heroes were only permitted to kiss their horses. We applaud the approach to reality, but then we are no longer a youth."

From the teletypes of *The Worldwide Press*:

"*Dusty Britton* today arrived in port, bearing a tale of a Galactic Civilization called *Marandis*. This Galactic Government it seems, intends to move the Earth to another sun because our position interferes with their program of running Galactic Highways back and forth across the trackless wastes of space. Moving Earth is a simple process, according to *Dusty Britton*. A mere matter of barytrine fields, machinus forces, phanoband carriers, and a general abandonment of the theory of general relativity.

"From the viewpoint of the scientists interviewed following this claim, *Dusty Britton* may or may not have been reading one of his own scripts. Knowing *Dusty Britton* of old, we are inclined to call this one: *Manuscript Found In A Bottle* with a deep nod at Edgar Allen Poe for the use of his title.

"Dr. Foster of the Wellmann Observatory suggested that enough of *Dusty Britton*'s story was logical to make it sound good. A race traversing the galaxy at hundreds of light-years per hour would find variable stars helpful if used as beacons. But Dr. Foster said that *Britton*'s story was illogically incomplete. If this outfit has the machinery necessary to move a planet, why not

move the stars themselves and create a straightaway passage from one end to the other without curves in the course?"

From The Wall Street Journal:

D' B' ttn Ent' pses-Open 68 Close 43 off 25

Editorial From *The Journal of Temperance*:

"Elsewhere on these pages is an apology for not printing the interview between our science reporter, Miss Agatha Westlake, and Mr. Dusty Britton. The interview was not concluded because Miss Westlake believed that she could detect the fumes of alcohol on Mr. Britton. It is deplorable that the youth of this fair land have put their faith and their future ideals into the character of a man of such despicable hidden leanings. A package of cigarettes was visible on the deck of Mr. Britton's boat and nearby was a small glass of the kind only found in those dens of iniquity, the formal name of which is forbidden to these pages.

"Let us therefore seek a new champion, who will eschew these vices; who will find it more godlike to extend his gracious invitation of vacation time to his youthful admirers instead of a woman of low moral fiber. We feel—"

TIME *Magazine*, Science Section:

"Dr. Willy Ley, in an interview today in his retirement home in Jackson Heights pointed out that he had always been convinced that the limiting value of the speed of light was a false theory. Therefore Dr. Ley concluded that it was entirely possible that an extra-solar race could have developed interstellar travel.

"My grandson, Gregory, is aboard the Venus Rocket," said Dr. Ley in the rich German accent that seventy five years in New York have not diluted. "I hope to see the day he takes off for Alpha Centauri.

"But I do feel that there is reason to doubt the story offered by Mr. Dusty Britton. Certainly the more intelligent persons of any galactic civilization would be less likely to contact an actor than scientists or government officials? This story of phanobands,

barytrine fields and menslators sounds too much like the fancies of science fiction to me."

Article in *The American Weekly*:

"With heat rays and weapons of unimaginable power the enemies of the Earth will swoop down to—"

From *The Chicago Tribune*:

"Not since the days of King George III has the threat of foreign entanglements been so great—"

From *The Daily Worker*:

"Without a doubt this advanced culture has developed a perfect galactic State, capable of serving all men according to their needs. We feel that a pardonable mistake has been made by their representatives in contacting a man of Dusty Britton's character, and we will wait with open arms the return of the galactic emissaries, who will bring with them the glories of—"

From Mount Palomar:

"Variable stars are of natural origin and can neither be started nor stopped. The theory that such stars are used by a galactic civilization as beacons and celestial stop-lights is utterly fantastic."

From the teletypes of *Worldwide Press*:

"Dusty Britton was arraigned today in Federal Court for having violated the rulings of the Federal Communications Commission and the international rulings of the Havana Conference of 1972. An indictment is expected from the grand jury, still in conference.

"Dusty Britton is charged with having caused the transmission of a false distress signal. He pleaded not guilty at his arraignment and will probably plead not guilty if his case comes to trial. A fine of ten thousand dollars or three years in jail (or both) is the maximum penalty for a conviction. Public sentiment will probably make the maximum sentence mandatory; this is an election year and the Administration is interested in

demonstrating that its foremost desire is to serve the public interest."

Press Release from Cosmic Studios:

"The filming of first run of the new series, *Jack Vandal, Space Rover* was completed here after an extensive eighteen day program. Jack Vandal is patterned after the characters of The Saint and The Lone Ranger. Unrestricted by the laws that prevent a policeman from performing his moral duty, hated by the underworld, Jack Vandal is to become a Robin Hood of Space. The world premiere will take place at The Palace Theatre, in Greater New York."

Statement from The Office of Scientific Research & Development:

"No evidence has ever been found to corroborate Dusty Britton's statements that radiation phenomena exist which cannot be explained by the application of Maxwell's Equations, and which are not subject to the limitations imposed by the theory of general relativity."

Ruling by the Bureau of Navigation, Marandanian Sector:

"It is hereby granted that a barytrine field be established about the Planet Three of Sol, and that Planet Three shall then be transported and placed in situ near a star of appropriate dimensions. This enactment is to take place at the convenience of the Transgalactic Company with the proviso that no inconvenience take place to the culture of Planet Three. It is ruled herewith that the change in stellar hemispheres and the revision in planetary pattern is of no prime importance to a primitive culture.

"It is further ruled that the loss of approximately one thousand years of direct time in the inhabitant's life is of no importance since contact with the external culture has not taken place, and therefore this loss has no bearing on the primitive culture. At the end of this period of transmittal, investigatory contact will be made to formulate a program of enlightenment which will result

in the eventual assimilation of Sol Three into the Grand Galactic Government.

Signed, Sealed, and Delivered
BuNav, by Direction."

V

Barbara Crandall opened the door for a quick glance, then opened it wide. "Oh. It's you!"

Dusty nodded glumly. "Yeah. Surprised?"

Barbara shrugged. "A bit. When did they let you out?"

"This morning."

"Rough?"

"You said it. Was it rough on you?"

"A little, but it's been made up for."

"How come?" asked Dusty looking up.

She smiled quietly. "I've got legs and a figure," she chuckled. "I've been cheesecaked all over town as the *Star Girl* and there's talk of my getting a part in the Jack Vandal series over at Cosmic Studios."

"How so? Seems to me that we're both sort of washed up."

Barbara shook her head. "Jack Vandal is a sort of cheerful villain, you know. He takes delight in bumping off the well-protected crook who can't be touched by the law. He's hunted by the police and hated by the underworld —"

"Spare the gruesome details. They haven't changed in a couple of thousand years. How come you're not in the dog house?"

Barbara smiled. "Because the woman in that kind of opus is always a sort of shady lady herself. It wouldn't do to have an innocent virgin for the

companion of a buccaneer. So with my slightly tarnished reputation I'm a natural. What happened to you?"

"The lie detector test."

Barbara blinked. "Then didn't that prove your point?"

"I thought it did. But I forgot one thing. Seems that the lie detector, no matter how good, is capable only of showing whether the character is telling a falsehood or not."

Barbara smiled confidently. "So you were telling the truth. Weren't you?"

"Sure," grunted Dusty. "Sure I was. But, quoting what's-his-name in the Bible: 'What is Truth?' One of the court psychologists pointed it out very clearly. If I firmly believe that the moon turned bright purple at ten o'clock last night, under a lie detector I'd be credited with a 'Truth' when I said so. In fact, the damned thing would say that I was telling a lie if I believed that the moon was purple and tried to cover up by saying that it hadn't changed. Follow?"

"So what was the verdict?"

"The verdict was to the effect that I was suffering under some hallucination—possibly induced by alcohol—which led me into this story. Therefore my lie-detector acquittal was valid only to prove that my call for help was, at the time, due to my personal conviction of danger. I was adjudged temporarily incompetent."

"What kind of sentence? They didn't just let you go."

"I've been two weeks in the observation ward of the federal looney locker. You see, to prove me guilty, they had to show that I had willfully and maliciously transmitted a false signal, with intent to deceive and/or for some personal reason. Willful tampering of this nature comes out as malicious mischief; malicious tampering becomes a federal offence. Maybe I've got my terms mixed up, but I think you get the idea, anyway. The end-up was this: Dusty Britton was convinced of his personal danger, his emission of a distress signal cannot be called malicious. I am no longer the top star I was once—in fact Gramer has cancelled my contract on the moral turpitude clause and the McDougall Office has black-balled me from all productions. So after a couple of weeks of observation at the spin-bin, they let me free with an admonition to leave the stuff alone. Barb, have you got a drink?"

"Sure thing. Look, Dusty, I know what you must think, but please don't ask me to corroborate your story. Not again."

Dusty nodded soberly. "I won't. The first time I thought we could convince 'em. But not any more, kid. One of us in the mud is enough. We've got to find a new attack."

Barbara handed Dusty a highball which he sipped before he said, "Barbara, we've got to do something."

"Why?"

He looked at her, stunned. "Why?" he cried.

Barbara took a sip of her own highball. "We won't lose a damned thing and you know it," she said quietly.

"A thousand years—"

"So what?" she asked simply. "Supposing that they were a bit more accurate than Scyth predicted. Suppose that they took this thousand years out of our life at a time when you weren't looking at the sun. Do you realize—" Barbara's voice lowered a bit dramatically, "—or have you been watching the night sky to see whether they have already?"

"I have," he admitted with rising excitement.

"All right," she replied complacently. "Then you surely must realize that this thousand years out of your life isn't going to change the stock market a point, or anything else."

Dusty nodded. "This I can realize. But do you think I like losing everything but my other shirt? Do you realize that as of this moment I've got only a couple of thousand bucks tucked away and about as much prospect of landing another job as a dead fly?"

"You're not really worried, are you, Dusty?"

"Why shouldn't I be?"

"Because as soon as this barytrine field goes on and off and we find ourselves around another sun, in another sky, you'll be corroborated."

He looked at her. "Of course—and I've kept my big trap shut, too."

"You've what?"

"You don't think I'd be nuts enough to go around telling people 'Well, if you don't believe me, just wait until next month!' do you?"

"Why not?"

"Because then they'd have carefully kept me on ice until after the big event."

"After which your story would be corroborated and you'd—"

"I'd have nothing," said Dusty sharply. "It's not good enough. Sure, I'd be corroborated, but then I'd be blamed for not being effectual enough to convince people in the first place. I'd be blamed for not being the guy I've been depicting on the stage. I've been Dusty Britton, The Great Hero. But when it comes down to really doing something, I'm Dusty Britton, Liar First Class. Next it is going to be Dusty Britton, Helpless Incompetent. I can't just fold my hands and tell 'em that they can wait and see, and then yelp 'I told you so!' because if there's anything that people hate it's 'I told you so!' characters."

Barbara Crandall looked at Dusty pityingly. "Dusty," she asked softly, "Just what do you hope to accomplish?"

"I hope I'll be able to—"

"No. I know what you want to do. But what I want to know is how."

"There must be some way—" his voice trailed off.

"I can't see it. Scyth has probably gone to Marandis to get his generator. Dusty, do you know where the hell is Marandis?"

"Somewhere towards the galactic center."

"I'm told that the galaxy is a hell of a big place. You've about as much chance of getting there as you have of swimming the Pacific Ocean with one arm tied behind you. Scyth is gone from here so far that it takes light thousands of years to get that far. Hell, Dusty, at this moment, the best resources of all the science of the Earth and the so-called planetary income couldn't move a housebrick from here to Venus in less than a matter of months. Alpha Centauri is actually no more than a dreamer's symbol so far as we're concerned. In fact, you and I know that Scyth's little friends are

somewhere on the dark side of Mercury getting ready to make Sol a variable. We couldn't get there for months and months, and then we'd have a hell of a time locating them, even if we had whatever it might take to get there."

Barbara thought for a minute and then went on, "And if we could direct the entire Earth, and could call upon anything or anyone, we wouldn't know where to start. What is a phanoband? Why is a barytrine field? Even I know that there are a couple of dozen rather brilliant men who believe that the speed of light is not a limiting velocity, but this is only a conviction, not founded on any experimental evidence. So maybe you've got a firm inner drive to go out and prove yourself. But how in the hell are you going to make headway against a race that considers us primitive?"

"We've got to make contact."

"How? Shall we call Mercury on the phanoband communicators? And what was that intermediary step? The machinus fields? It sounds like double-talk to me."

"It was something about abandoning general relativity for the machinus theory of space-time," said Dusty, bringing into focus all the science fiction he had ever read.

"Got any theories?" asked Barbara pointedly. "Frankly, Dusty, I'd like to help, but I feel too much like a man trying to come all the way from the stone age to the atom bomb in ten days. In order to circumvent their foul plan we've got to abandon a very workable theory in favor of an unknown something called the machinus theory of space-time, and then from that we develop something called phanoband radiation, which produces factors enabling us to reduce the theory to practise and eventually we take to deep space, find Marandis, and put our case in front of some sort of bureaucratic something-or-other. Can't see it, Dusty."

"So what am I supposed to do?"

"Sit and take it. What else can you do? Darn it, Dusty, you can't fight them, and you aren't in any position to join them. We haven't got the initiation fee, we don't have the address, and we hardly talk the language."

Dusty looked at her sourly. "I'd hoped you'd help," he said unhappily. "You at least know what the score is."

"Dusty, I'd like to help. I do know what the score is. It's hopeless. You're trapped in an awkward position. And like a lot of other people, you are in a position where you can't do a damned thing about it. So you might as well save your high blood pressure and start looking around to see what you can make out of it."

Dusty finished his drink and left. In a trash-can by the alley was a Dusty Britton Blaster, complete with holster and a tin medal for sharpshooting. The school-store across the street was displaying a Jack Vandal mask and a small case containing ten candy cigarettes and a secret compartment suitable for concealing ten-thousand dollar bills lifted from lawless characters who might have used the dough to bribe juries or buy professional gunmen.

Dusty made his way along the street unrecognized.

The guard at the front gate looked at Dusty with suspicion. Dusty looked back defiantly; for a number of years the guard had practically bowed thrice as Dusty approached, Dusty hoped that the habit of deference was well established.

"Have you a pass, Mr. Britton?"

"Now see here, Sam, I don't need a pass and—"

"Mr. Britton, I've got orders to—"

"Look Sam. Let's not stall. I want in and I'm going to—"

"One minute, Mr. Britton. I'll have to call."

Dusty grunted. "I want to see Doctor Ross."

"Oh. Well, just a minute."

The guard called, and Dusty could hear the roar of Martin Gramer, "Throw the louse out!"

"Sorry, Mr. Britton. We can't let you in."

"Look, Sam, I've got trouble. You've got trouble. Do you remember your younger days, Sam? When you were the top boy at Graphic Arts?"

"Sure do. Great days, too."

"What happened, Sam?"

The smile faded from Sam's face. "I got too old."

"Sam, all I want is to gab with Dr. Ross for a minute or two. I've got a great idea. And I'll make you a promise, Sam."

"Promise?"

"Sure. I'll promise you that if you let me in right now, and this idea of mine goes through, that I'll see that you get a good bit in anything I'm in. We'll work it up from character actor until you're playing bigger and bigger bits. You can make a comeback, Sam, and I'll help you then if you help me now. How's about it?"

Sam looked through the studio gates for a moment, and the thinking could almost be seen in operation. He had darned little to lose; he could always blame Dusty's entrance on some dreamed-up excuse, and if Dusty's idea worked, he might even be able to take credit for having used some initiative.

"It's a deal, Mr. Britton. But don't forget me."

"I won't."

Dusty went inside, found the main idea-office, and talked himself into the office of Dr. Ross. These hurdles he found less difficult than the front gate; possibly due to the fact that once a man was inside the fence, everyone thought he belonged there.

Doctor Harold Ross greeted Dusty with surprise.

"Dusty! How goes it?"

"Not good. I'm a professional louse."

"How come?"

"Don't you read? Forget it. Look, Doc, you're actually the only scientist I know, so I want to ask a couple of questions."

"I'll try. But let's not lose sight of the fact that I'm not a credited scientist, as you put it. I'm a sort of cockeyed physicist whose job is to see that actors squinting through telescopes see Saturn at the right angle, and that birds looking through spectroscopes don't point at a blue triplet and call it the Sodium D Lines."

"You might be even better than a real physicist of the research kind," said Dusty.

"Thanks for them kind words, Dusty. Flattery will get you nowhere."

"I'm not trying flattery. You've been in this make-believe business for a long time. That's why you might be able to think it out."

"Go on, man. Spill your idea. What do you want me to do?"

"Let's assume that Dusty Britton's wild tale about a man named Scyth Radnor, from Marandis, is right. And that this guy came out of a spacecraft parked in the ocean, sitting on the sill of the spacelock waiting for me. He talked about the death of the general relativity theory in favor of something called the machinus theory of space-time, phanobands, menslators and all sorts of things."

"Yeah? We've been having space warps ever since the days of Jack Williamson."

Dusty grinned, perhaps for the first time in weeks. "Look," he said. "I know the patter well enough. Doc Smith invented the Bergenholm and Murray Leinster came along with the superdrive and George O. Smith developed the matter transmitter to a fare-thee-well, but all this guff is so much birdfood."

"What are you getting at, Dusty?"

"I wish I had studied a bit more science," said Dusty plaintively. "But dammit, I don't know a microfarad from a polysyllabic neutron. But I'm telling you that my so-called strange fancy is the God's Truth. Some time in the next couple of weeks the Earth is going to get itself transplanted. You can either help me now or you can come back later and tell me that you're damned sorry you tossed me out. Take it or leave it."

"All right. So maybe I'll take it. I've only a couple of weeks to lose. What do you want me to say?"

"Look, Doc, supposing that you were convinced that interstellar travel is possible; that these phanobands do exist. That this menslator is a commercial instrument. And so on. Take the first premise: faster-than-light travel is a commercial fact due to the development of a theory called the machinus theory of space-time. Can you do a bit of hypothetical theorization?"

"Sure thing. I don't mind. We'll take this on the basis of plenic syllogistics. Our first premise will be that this menslator works as your pal Scyth claims."

"It's Scyth. Not scythe."

"Then as I put it, the menslator produces the mental image that Scyth intends. He will say, for instance: 'A gostak distims the doshes,' and because he means that a professional preparer of comestibles has placed an unstated number of crustaceans under an open flame, you receive this statement of Scyth as: 'The cook broiled some lobsters.' Is that clear?"

"I can follow you," said Dusty. "This much Scyth explained."

"Good. Now let's look at our commonly accepted definition of 'Mechanus'. This means that it works. In other words we have him telling us that their culture has developed a 'workable theory of space-time' which has been taken up after the theory of general relativity displayed a number of gaping holes. So their 'mechanus theory of space-time' is a workable theory."

"And where does this lead us?" asked Dusty.

"Right back into a circle," said Dr. Ross thoughtfully. "Because if they've developed interstellar travel due to considerations brought about by the mechanus theory, that means that they have proved their theory by practise."

Dusty grunted half-humorously. "Isn't this like saying that mud is sticky because it's gooey? Or that winter is cold because of a lack of heat?"

Ross nodded. "Or that things fall because of the law of gravity."

"But aren't all these things a case of defining 'A' in terms of 'A'?"

"What isn't?" demanded Dr. Ross. "You're not looking for the Universal Truth, are you?"

"No, but—"

"Look, Dusty, the reason that we can afford to accept the fact that one and one adds up to two is simply due to the fact that one and one adds up to two in a great majority of cases."

"Wait a minute, Doc. One and one is always two."

"Not when you add a quart of alcohol to a quart of water. One and one here adds up to about one point eight."

Dusty waved a hand. "That's different."

"Not by a long shot, Dusty. There are extenuating circumstances. But this is just a proof of the fact that one and one is not always two."

"All right. But where does this leave us?"

"In the same damned circle. Granting that your observations are correct, proper, and unwarped by the addition of bourbon, Scyth and his galactic civilization have developed faster-than-light travel which has resulted in the establishment of a galactic government. But the explanation of how it is done cannot be derived from the nomenclature of the theory. Frankly, I have not the faintest idea of how to go about unravelling the word 'phanoband' unless we take it apart from its roots. Let's see, now."

Brows furrowed and lips pursed, the physicist thought for a long time and then looked apologetically at Dusty.

"I may be off the beam, Dusty, but I have a notion that your own mind put it together this way: Phan probably pertains to the roots of phantom, or unreal, or ghostly, or what is commonly referred to as the 'supernatural.' The so-called supernatural is invariably a phenomenon which cannot be explained by commonly accepted academic theory or empirical practise, mostly because the folks who work with it have neither academic nor empirical data. Incidentally, the 'o' part of this first phase is undoubtedly a conjunctive vowel stuffed into the word so that it can be uttered without losing a couple of front teeth or blowing a vocal fuse, or maybe spraying the listener like a professional German lecturer. So let's accept the concept of 'Phan' as something that you cannot explain in common terms."

"Go on, Doc. You're reducing my case to an absurdity, you know."

"I'm sorry, Dusty, but that's how I see it. Now, let's take the 'Band' part of the word. As a disciple of Maxwell, et al, I am hopelessly incapable of concocting a workable theory of radiation which has nothing to do with some basic concept of frequency. Frequency, when you sit down and start analyzing it, is a nice, stable idea that explains a hell of a lot, Dusty, and as you get into atomics you find that particle radiation can be mathematically reduced to terms of frequency. You can actually compute the equivalent frequency of a thrown baseball or a .22 rifle bullet, you know. Then we get to that high-flung miracle we call 'resonance' and God protect me from having to deliver a thirty-minute explanation of resonance."

"I won't ask you to, Doc. But aren't you getting involved in your own traps?"

"Yes, I am. And I'm sorry. But I can't help it. But you can follow my fumbblings, Dusty. In the first place the radiation is not understood, which explains your accepting the mental concept as 'Phano' and because the physics of the radiation must be other than electromagnetic—which would call for the menslation into 'spectrum' the somewhat ambiguous term 'band' is assigned in your mental concept of the idea. So the literal menslation of the word is: 'Unknown mode of radiation' which—"

"But where are we getting, Doc?"

"That's what I was approaching, Dusty. This harangue boils down to the following: these people have a form or type of energy level which is completely inexplicable to terrestrial science at the present state of the art. Their terms, when menslated into our level of appreciation, come out as 'something that works' and 'something that cannot be defined' which, after all, is like trying to explain to a savage why a hunk of black rock always turns toward one direction."

"Hell!"

The doctor continued. "Sure. It's hell. Even your own term 'menslator' which I've picked up as a fine concept is only your own feeble transliteration of the definition. It does not carry any of the basic theory. So the fantastic gizmo

merely aids in the conveying of an idea from one mind to another, despite the fact that the two minds place different values upon the definition of words."

"But this isn't what I'm getting at, Doc. What I want to know is: granting the possibility of faster-than-light velocities, what have we got to explain it?"

"Nothing. Nothing but your own statements that you believe that this is possible and that someone has done it. None of us have any evidence that it is possible, except you. And I am afraid that I must question your training as a scientific observer."

"But, Doc, I—"

"Let's face it, Dusty. You swing about as much weight in scientific circles as Suzy Richtmeyer, voted last year as Miss Alphanon, parked on the Caltech boo-hucky showing about three yards of shapely nylon and thirty-two well-polished teeth. She was gorgeous but ill-educated, Dusty. She wasn't afraid of getting sterile in a radiation lab. She was afraid of getting pregnant. But if you sit there and ask me how anybody could possibly make any sound and workable theory out of what you describe, I can't see it."

"Look, Doc, maybe I can't deliver much. But they were there and that's what the guy told me."

"There's only one hope, Dusty."

Dusty Britton looked at Dr. Ross; with a voice of determination he said, "Doc, if there's any hope, let me know how?"

"You've claimed that this galactic gang have some humanitarian instincts. They aren't just going to set fire to good old Sol and let us alternately fry and freeze."

"Stop kidding me."

"Maybe I'm not kidding. I'm still promulgating on your own cockeyed plenum."

"You're not giving me much—"

Dr. Ross sat back confidently. "No, dammit, I can't say that I give much credit to your cockeyed story, Dusty."

"Now see here—"

"Now *you* see here," snapped the physicist sternly, "I won't deny that anything is possible. But I am a firm believer in the law of least reaction, and I think that this covers the case. If this character Scyth is at all concerned about our welfare—still granting that he does exist elsewhere but in your own mind—then get this, Dusty Britton: he will be back to see how you've made out in your program of preparing people for the big change before he turns on this barytrine generator."

Dusty eyed Dr. Ross sourly. "And what is your explanation of that word?"

"Easy, and it means no more than anything else when it is what you call menslated. 'Bary' stems from the root 'heavy' as in 'barytone' referring to something of heavy voice or highly accented. 'Trine' refers to something threefold in astronomical or—er—astrological (haruumpf) meaning. My God, Dusty, the word itself pertains to something as three-times-as-heavy. You don't expect me—or any other scientist—to come up with something sensible from a bunch of half-baked definitions, do you? All you've given me so far is a workable theory, an unknown medium of radiation, and something that is three-times-heavy. Tell you what, chum. Bring me your Scyth Radnor and introduce me. I know guys who would analyze MacBeth's three witches' brew if they could get a microgram sample. But not from that gobble-gabble about the 'fillet of a fenny snake, in the cauldron boil & bake!' line out of Shakespeare." The physicist went on in an undertone, "Eye of frog and tongue of newt," until Dusty stood up and prepared to leave.

VI

Scyth Radnor was pleased with himself. The trip had gone well. He was back on Earth and the barytrine generator was running in the warm-up cycle, building its field to the magnitude necessary for synchronization to the fabric of space stress caused by the planet Earth. It had not been difficult to maneuver himself into this position of having to run the barytrine generator and in doing so turn up with a few days of vacation.

He surveyed himself in the mirror and nodded. Then he left the big spacecraft and embarked on an errand that looked very interesting indeed.

Eventually, with no adventure worth reporting, Scyth found himself standing before a door pressing on a button.

Barbara Crandall cracked the door an inch or so and peered out. "Yes?" she asked. Barbara was not expecting any visitors, and her natural reaction was to open the door only a few inches until she determined the person making the call. But the sight of this man in faultless whites caused her to open the door a full two feet.

"Miss Crandall, I—"

"I don't think I—"

Scyth chuckled again. "Barbara, may I call you Barbara?"

"Oh, now see here—"

"You don't know me?" demanded Scyth with a hurt expression.

"Should I?"

Barbara was beginning to doubt this parley as a program of good sense. As a stage personality, even though far from a universal popularity, she knew very well that a completely dull heart frequently beat lustily beneath an expensive exterior and that a clear, open, friendly face often went with a mind fit only for the company of scorpions.

He saw her doubt and decided that he had played this guessing game long enough. "Barbara Crandall, I know you don't recognize me in these clothes and in this surrounding. Our last meeting was under a rather strange circumstance. I am Scyth Radnor, the Marandanian."

"Scyth Radnor!" she exclaimed. "I—yes, it is. I'm sorry, Scyth. I did not recognize you in human clothing."

"Please," he parried, "Don't say it that way. I am as human as you are."

Barbara looked at him defensively. "And you're here to prove it?"

Scyth blinked. She was rather distractingly direct. "There is no suitable answer to that," he said. "Must I supply one?"

Barbara laughed. "Come in, Scyth. Let me offer you the hospitality of a drink."

"Pleased," he said, following her into the living room. She waved him into a chair and turned towards the kitchen.

When she came back with two highballs, Scyth was relaxed in the loveseat. Barbara noted it with inward amusement and handed him the drink without comment. Scyth sipped the drink first and then took a deep and appreciative drink.

"You do have something to offer," he said, not showing his disappointment that Barbara had seated herself in the chair instead of on the loveseat beside him.

"That," she said, "makes two items, doesn't it, Scyth?"

Scyth knew that he had lost the initiative; Barbara was way ahead of him. He tried another tack:

"I came to see how you are making out," he said.

"I'm not doing badly."

"Is the public aware of the impending event?"

"Aware, but not believing. Dusty Britton lost his shirt over this."

"He'll get it back," said Scyth. "I'm not concerned over the result. It's happened before and it will probably happen again."

"It's more than possible that Dusty will be vindicated but will then be blamed for not doing something about it," said Barbara.

"That cannot be helped. Dusty couldn't do anything about it, you know. And if Dusty loses out in the long run, we can't permit the well-being of one lonely man to stand in the way of galactic progress."

Barbara smiled confidently, but with a slightly sour twist to her pretty lips; it led Scyth to think that there was some derision in her mind. She confirmed it by saying, "Scyth, since you are going on with your program no matter what happens, and your concern about warning the people has worked no matter what happens to Dusty Britton, why do you bother coming back for a look-see?"

Scyth squirmed uncomfortably. Despite certain jokes to the contrary, it is not acceptable to confront a desirable young lady of barely speaking acquaintance and flatly state the delicate proposition. The difficulty here was that no matter how he tried, Barbara Crandall was turning the trend of conversation right back onto the old original trail.

"You're an actress," he said.

"So I'm told."

Scyth smiled. "You're popular? You are in demand here?"

"I am on my way up," she said.

"Barbara, you could be a popular actress, you know."

"Someday I shall be. But this does not come overnight, Scyth. It takes work, you know."

"I have an idea that the flavor of the foreign often helps."

"This is true."

"Then I have a suggestion. Why not come along with us back to Marandis? You have youth and beauty and ability and also the exotic flavor. It—"

"What shall I be?" she returned quietly. "The ignorant but beautiful barbarian? A clothes horse slightly incapable of holding an intelligent conversation? This seldom works, Scyth. I've studied history a bit and I recall the case of a native girl called Pocahontas who was carried from her native surroundings into the height of the civilization for the time. She was no actress—she was *exhibited* like a pet monkey or a rare zoölogical specimen. She died of what they called heartbreak. I think heartbreak in this case was a combination of loneliness, of facing the realization that she could never really belong to the culture, of the futility of asking to be returned to her people. In other words Pocahontas lost the will to live. So thank you, Scyth, but I have no desire to be a chattel, or a curiosity.... Or a museum-piece."

Scyth nodded seriously. "I see your point. But I don't agree with you. In the first place you are indulging in a conversation with me. In the second place, you—"

"In the first place," said Barbara pointedly, "this conversation is being carefully kept on my level, isn't it?"

"I wouldn't say that."

"Of course not. But look, Scyth, aren't you using that menslator of yours?"

"Of course."

"Then the menslator keeps the conversation down to my level because by its very nature it cannot convey an idea to me that is beyond my understanding. Am I correct?"

"In a sense, yes. But—"

"Scyth, can you menslate a dog, for instance?"

"A dog has so little mind that—"

Barbara interrupted this with a wave of her hand. "So how long would it be before you and your people became damned sick and tired of talking down? It would be like trying to conduct an adult discussion in baby talk, wouldn't it?"

Scyth shook his head. "Not entirely," he said. "It might be that way at first. But this would not last. I don't know of your history, but I assume that your Pocahontas was a true savage. You had nothing like the menslator. Doubtless she never learned any real language and so lacked the ability to use a language of any kind, let alone learn the ramifications of the culture behind it. You would be on an entirely different plane. You have a language and a culture and you are quick to grasp a new idea. With a menslator you would learn the language well enough in a short time and while the deeper factors of the culture would always escape you, the superficial parts would eventually come easy."

For an answer, Barbara pointed to the wall. "Scyth, on that wall is a painting given to me by a character who calls himself an artist. Take a gander."

Scyth looked. The painting was a mess of squiggles and blots of color. It was iridescent here and drab there, soft lines elsewhere and sharp contrasts

somewhere else.

"Interesting," said Scyth. "What is it?"

"I'm not sure. I think that this is the painting; but all it needs is a hole in one corner and it could be the palette that the guy used to make the painting."

"This is apropos of what?"

"Frankly, I think it is a mess. It is something that could be accomplished by a monkey turned loose in a paint store. But the artist calls it 'modern' and defends his stand by stating that anybody who criticises it is wayward, ignorant and unappreciative of the finer moods and things of life. So put me in your culture and turn me loose. If I criticise it will be because I am too primitive to understand these higher bits of culture. If I enjoy something, I am looked down upon because I can't really feel the true depth of the thing. It—"

Scyth held up a hand and his empty glass at the same time. Barbara laughed and went to give him a refill. It also gave him time to think, and when she came back with his highball he had the answer.

"Barbara," he said sincerely, "a lot of what you say is true. But look at it this way. You will be a celebrity. You will, to all intents and purposes, be among your own kind. That helps. So you can't follow the deeper arguments nor appreciate the complexities of society as we know them. But think of what you can see and enjoy which will be forever denied you if you refuse my offer."

"For instance?"

"Imagine the beauty of a planet under a double sun. Imagine if you can the beauty of a night sky with a ringed moon glowing soft over the landscape. Coalestis is a planet where most of the minerals and rocks combine into black stuff. Imagine the beauty of a city of polished ebony. There are the twinworlds we call Venago One and Two. The Venagos are separated only by about a hundred thousand miles and in the night sky you can look up and see the other world glowing over a quarter of the heaven, and on the dark side are the winking beauties of the cities glowing like jewels. You will see worlds where the vegetation grows lush; riotous colors to hundreds of feet tall and there are cold planets where the ice and snow are always dazzling white. You will wear sheer shimmering cloth so soft that you have no word

to describe it. You will wear jewels that glow with their own internal light. Money and luxury will be yours, to travel as you see fit; to spend the rest of your life flitting from star to star, seeing the varied wonders of the universe. That is the fate of an actress in our culture, Barbara, for Lord knows we have few enough of them."

Barbara looked at Scyth seriously. A number of things occurred to her, and one of them was simple. If Scyth had returned to earth to see her, it was obvious that she measured up well against the women of Marandis. Another factor was the yearning to travel. Barbara would not have recognized the train of thought if it had been labelled and explained, but it was there none the less. This was her one chance to see the greener grass on the other side of the galaxy, the chance to realize a human dream of countless centuries.

She smiled wanly.

"You see what I mean?" asked Scyth.

"I think I do."

"Doubts?"

"Yes. I feel as though I'll be abandoning my own kind."

Scyth had been leaning forward on the loveseat. Now he came forward to cross the room. He leaned down, took her hands, and lifted her out of her chair.

"You'll come?"

"You make it very attractive."

"You can do nothing by staying, Barbara."

"But—"

Scyth freed one hand and fished in his jacket pocket. He came up with a small box, deftly flipping the cover up with his thumbnail.

Coiled inside the box was a chain of tiny-linked metal that glowed gently with a pale green light. Against the dark cloth of the box lining was a scrollwork of dark metal, the setting for a stone about a half inch in diameter.

The stone itself was cut in many facets each of which glowed in a dazzle of a different color. Scyth moved the box gently and the facets changed color and sent flecks of polychrome dancing against the ceiling, the walls, the floor. Flecks of light caressed his face and sparkled into her eyes.

Barbara took a deep breath, then held it, completely entranced by the bauble for which she had no words to describe. It was sheer beauty and she knew that anything that she said would be completely inadequate.

Scyth freed his other hand and took the pendant by the chain. Holding it by both ends, he held it up to her throat.

Barbara stood immobile as Scyth put his hands to the back of her neck and fastened the clasp. Deliberately he let the tiny links slide down across her shoulders, let the chill of the cold jewel-stone thrill her as it slipped down her chest towards the hollow between her breasts.

Then, gently, Scyth took her by the shoulders and turned her to face the mirror on the door. She turned under his hands as though she had no will of her own, to look into the mirror and gasp at the rich beauty of the gem.

Scyth drew her back against him and she leaned gently with her forehead against his chin. He put his hands on her waist and she covered them with hers, squeezing them as she drew his arms close around her. She tilted her head back and turned her face to offer her lips and he found them warm and soft. His hands caressed her. Barbara turned in his arms to face him and he held her close.

VII

The snick of a key in the lock did not break through their preoccupation with one another, but the cynical voice of Dusty Britton came as the shock of a bucket of cold water:

"Very pleasant scene," he drawled. "I hope I've interrupted something."

Scyth and Barbara parted in a whirl.

Scyth felt a sinking sensation in his middle as he realized that the facts were far too clear; that the sensible course was a hasty retreat, but the only path

was barred by Dusty Britton.

Barbara took the woman's course. "Don't you ever use the doorbell?" she asked icily.

Dusty smiled sourly. "I always have," he said. "Up to now. But this time I want words with the gentleman in question instead of losing him out through the back door."

"I think I should explain," said Scyth uncertainly.

Dusty chuckled. "What sort of explanation do you think I'll accept?" he asked the Marandanian.

"But I—"

"Stow it, Scyth. You couldn't explain a thing and you know it."

Barbara snorted angrily. "See here, Dusty, you can't come in here and start —"

"I'm not starting anything. I'm just seeking a conference with Scyth."

"How did you know?" asked the Marandanian uncertainly.

"By being just smart enough to find a tomcat by knowing where the tomcat is likely to prowl."

"Meaning?" demanded Barbara icily.

Dusty ignored her. To Scyth he said, "I don't know beans about barytrine fields or generators, but I guessed that you'd set it up on earth somewhere, start it cooking, and wetnurse it until it came to a boil. That would leave you on Earth with time to kill. Since time hangs heavy, you'd probably look up one of the only two people you know. The more attractive one, Scyth. So I've been haunting the front door like a private eye."

Barbara coughed. "You took that right out of *The Space Patrol Fights The Overlords of Delgon*."

"So I've got good writers," grinned Dusty.

"What do you intend to do?" asked Scyth nervously.

Dusty faced Scyth. Dusty topped the Marandanian by perhaps an inch or two and covered him by a good twenty pounds. He guessed that if it came to

roughhouse he would probably win. He poised himself on the balls of his feet, just in case. He had no way of guessing the speed or power of the wiry-looking Scyth Radnor and so he was taking no chances.

"I became a professional bum because of you and your phanobands and your menslators and your barytrine fields," he said bluntly. "I was laughed out of everything I had. So now you're going to go with me and tell 'em all that I was right. We'll have the big domes out to take a look at your spacecraft, have 'em inspect your barytrine doodad, take a gander at whatever it is you call phanobands, and so on."

Scyth understood all too well. He was trapped, faced by a man who could take him apart bit by bit without much trouble, and if he came out of it alive, he would end up by being a bigger bum than Dusty Britton had become. Scyth had fumbled badly by taking time off for fun and games with Barbara and he knew it. The only thing to do was to clear out of here no matter what happened afterwards. For once the barytrine field snapped on, any evidence of Scyth Radnor's attempt at dalliance could not come to light for a thousand years.

His hand lifted slowly to the inside pocket of his jacket as he said, "I'll be glad to help you, Dusty. Naturally, none of us have any notion of making things tough for anybody. So—"

Scyth went into whirlwind motion. His hand came out from inside the coat carrying a fluted-barrelled weapon. As the end of the thing cleared the lapel of Scyth's jacket he was fingering the trigger and a pale emanence seared out and cut down and over in a slashing arc.

But at the whirl of action, Dusty's hand arrowed into the space between the lower two buttons of his dress shirt and came out with a snub-nosed automatic.

The pale slash of Scyth's weapon was blotted out by the flash and racket of a shot.

Scyth whirled, flinging his weapon against the wall from an outstretched hand. The thing hit with a crunching sound and Scyth continued to turn on

rubbery legs, sinking and sinking and turning until he sat heavily on the floor. He sat, stunned, just long enough to fold his hands over his belly. Then he folded forward over them and rolled around sidewise as if falling out of his own lap. He half-rolled and fell a-sprawl on his face. A spread of blood stained the white carpet.

Dusty looked down at Scyth. He looked from Scyth to the snub-nosed gun in his hand and swallowed heavily. The gun dropped to the floor with a muffled thud from nerveless fingers; Dusty looked at Barbara out of far-away eyes and said, "He—er—I—"

Then he slid to the floor in a dead faint.

Barbara stifled a scream. The whole thing had been lightning-fast, but she had caught most of it. Scyth had shot first but now he was bleeding on her carpet. Dusty had shot second and was lying in a dead faint. Hysteria choked up in her but she drove it back. She wanted to laugh hysterically. She wanted to let go and slide to the floor and go to sleep while someone else came in and cleaned up the mess.

Realizing that she could only hold off the rising hysteria until someone did make a rational move, Barbara reached for and drained the highball on the bar. She augmented this slug with a muscle-sized hooker from the bottle. The liquor burned down and helped to iron out her jittery nerves.

She grabbed the ice-pitcher which was filled now with melted cubes and a slosh of water. Unceremoniously she poured the cold mess over Dusty's white face.

Dusty's eyes fluttered and his voice made spluttering noises. "Wha—?" he fumbled.

"Come off it!" snapped Barbara.

Dusty sat up weakly. He looked around for a moment as if he weren't quite sure of where he was. Then he caught sight of Scyth and it all came back to him. He scrambled to his feet and took the bottle from Barbara's hand. He took a healthy slug himself and then said, "He tried to—tried to—"

Barbara laughed hysterically. Between gales of half-mad laughter, she said, "Tried to beat the fastest man—in The Space Patrol—to the draw!"

Dusty slapped her across the face with the flat of his hand. "Shut up!" he roared. "Shut up and make sense!"

She came out of the hysteria instantly, shrinking back from Dusty with a hand against the growing redness on her face. "Dusty—don't—"

He shook his head hard. "Sorry. You needed it."

"I know. But he—? Look, Dusty, what do we do now?"

Dusty looked down at the bleeding man. "Cops," he said thickly. "I've just shot a—" He could not finish; his face was turning green again.

"Cops nothing," snapped Barbara.

"But shooting—"

"Come off it, Dusty. The cops will only delay and investigate and generally botch things up until it will be two months and a thousand years from here."

"Cops aren't that stupid."

"Cops aren't stupid at all," she snapped. "They're just smart enough to insist on knowing all the answers. So tell you what. You go to the phone and call Lieutenant Yonkers and explain carefully that you've just shot a Marandanian Marauder in my living room. Tell him you've collected one of your Great Galactics, only he's defunct. See how far you'll get!"

Dusty looked at her blankly.

"The first stop will be the bull pen," she went on hotly. "The second stop is the nut-locker. And the third stop is some unknown star a thousand years from now while the F.B.I. try to match the guy's fingerprints. Then you call on me for a witness and that gets us the front page in big black letters saying: 'Former Hero Shoots Rival In Leading Lady's Boudoir!' Start thinking right, Dusty Britton. Or," she added scathingly, "call up one of your writers."

Dusty considered. "I could slope out of here and—"

"Like hell you will!" she screamed. "You're not leaving me here with a body to explain."

"But defending your—"

Barbara's scorn was high. "Look, Dusty, ever since we were sighted off-shore in the Buccaneer I haven't had a shred of virtue and everybody knows

it."

"Trouble is that we can't even run," grumbled Dusty. "This is your apartment."

Barbara looked down at Scyth. "Damned nuisance," she said.

The damned nuisance groaned. The sound was hollow and weak but it seemed to ring through the room like the cry of a wailing ghost.

Barbara cried: "He's alive—"

"—not dead!" blurted Dusty. "Get water and stuff."

Slowly they stretched Scyth out on his back, and Barbara went for her first aid kit while Dusty slid off Scyth's jacket and ripped the shirt free. The wound looked frightful, but some sponging with hot water and alcohol reduced the horror to a weeping hole that tried to breathe blood in and out. It was low on one side, somewhere near the floating ribs on the right.

"Flesh wound?" asked Dusty hopefully.

"I wouldn't know. Maybe." Barbara flipped the pages of a large book from her library, a book that had not been used much. "It says a compress."

Dusty made a pad of bandage and cotton and covered the hole. He taped it down. Scyth groaned again and Barbara cracked open an inhalant vial and put the stuff under Scyth's nose.

"Wh—wha—di' you hi' me wi'?"

Dusty never knew from where he found the moral strength to be hard-boiled. But all of a sudden the feeling that this was one hell of a mess left him; his next feeling was one of confidence and self-justification. "It's called a belly gun," he said. "But you'll be all right in a couple of months. Maybe three."

Scyth tried to struggle up but failed. He fell back and lay there glaring at them. He gasped, "Cou'le munce?"

"Sure. Stop crying. It's just a flesh wound."

"Bu' in cou'le munce—'ll be—bar'rine fiel'—gone—"

"Take it, Scyth. Sure. It's tough," said Dusty in a cold, matter-of-fact voice. "You've played and lost, but that's all right. Be a good loser. You've got a lot of company."

"Com'any?"

"Sure. There's millions of guys who've lost their future and their birthright over the flick of a hemline. We're a primitive sort of race, old man, but you'll find us both healthy and lusty. Forget Marandis and your ding-busted beacons. Maybe you can help us build a spacecraft—after we get through this barytrine business your friends cooked up for us."

"Bu' can—mus' not—Chat an' Bren—die—"

"Nonsense."

Barbara plucked at Dusty's sleeve. "He's talking about his friends. Chat and Bren. On Mercury, remember?"

"Oh, don't worry about them."

"But don't you see, Dusty? If we go into the barytrine field, and trap Scyth and his spacecraft with us, his friends will be marooned on Mercury."

Dusty nodded quickly. "Sure and that's what I'm counting on. They'll not start Sol into a variable until Scyth gets back. So—"

"Don't be blind. They won't start the variable star, but no one can stop the barytrine field. They'll still be marooned."

Dusty grinned. "You don't think a gang this advanced would be so dumb as to leave a couple of their kind marooned on a place like Mercury, do you? Well, I'll tell you how I've got it figured, Barb. Exactly eight seconds after Scyth does not land as per schedule, Chat and Bren will be calling for help on these phanoband things. That'll take care of them. But as for this guy, let's cheer up. We've got a sort of hostage. Scyth will be most happy to make a spacecraft for us as soon as he gets back on his feet. Chat and Bren will, of course, be taken care of some thousand years before we—"

Scyth groaned loudly.

"Huh?" demanded Dusty.

"S'no-so. Bren an' Chat—alone. No—no—famban—phan'ban'—phanoban' on Mer'cry. Die—"

Barbara started to say, "But your company—" but Dusty turned quickly and slapped a broad hand over her mouth.

"Shut up," he whispered in her ear swiftly. "He's got to think there is no help. He's forgotten that someone knows they're here. Play it by ear and follow my lead."

"What can you hope to do?"

"I don't know," said Dusty. "But I'm hoping that I find out." Loud enough for Scyth to hear, Dusty asked, helplessly, "But what can we do?"

"Car—ou'side. Spacer. Pocket—map."

Dusty made a dive for Scyth's jacket and found a folded road map in one of the pockets. Like any stranger in a strange land, Scyth had outlined the route in a heavy blue pencil. His travel was detailed, it took Dusty no more than a glance to place the location of Scyth's big spacecraft.

Scyth rested a moment and then went on: "Hurt—can be doc'or on Maran'is. Hurry—"

Dusty grunted. "And who's going to run this spacecraft of yours?"

"You—easy—"

Barbara looked at Dusty cynically. "It's your show, Spaceman Officer." She laughed hysterically again. "Dusty Britton Rides Again!"

Dusty slapped her across the face to shock her out of it. Then he bent down to look at Scyth. The compress was soaked with red blood, but it was not overflowing. Dusty touched it gently and looked up at Scyth's face. "Hurt?" he asked.

"Can' tell. Hur' all over."

"Gonna hurt more, Scyth. C'mon, make a break."

Dusty put his arm under the Marandanian's shoulder and slowly lifted him to a sitting position. The man groaned and the compress broke out in a new flood that ran wet for a moment and then subsided in the stickiness of clot.

Dusty lifted Scyth as gently as he could, and with Barbara opening doors, he carried Scyth to his big car.

"Why not take his?" she asked.

"Like mine better," he said with a shake of his head at the rental-agency model Scyth had come in.

Barbara found blankets from the trunk and made a soft cushion for Scyth.

"You take care of him and I'll drive," said Dusty.

Barbara shook her head. "I—you take care of him and I'll drive."

"But I know the route."

"I can read a map as well as you can."

Scyth opened his eyes wearily, but with a trace of bitter humor he managed to say, "You take care—of one another—and I'll drive!"

Then Scyth passed out cold.

Four hours' drive into the foothills, far from the lights of civilization, Dusty found the big spacecraft. It was parked in a small valley and it was colored so that only a man who knew what he was seeking and where it was would have found it.

On the way Scyth babbled about the drive and how to run the big ship. Happily, Scyth's periods of delirium were easy to separate from his periods of lucidity, for when Scyth began to babble he talked cynically about the stupidity of taking four hours to travel less than a couple of hundred miles when they could cover light-years in the matter of minutes. Then he would become quite rational and tell Dusty how to recognize the beacons as they came into sight, and where the charts were. He had to get back to Marandis, and he told Dusty the way.

Then his mind would wander a bit and Scyth would chuckle quietly over something entirely removed from spacemanship. Then would come a discussion of the levers that must be turned and the meters that must be watched; how to turn the correct knob or to push the proper pedal. He spoke

of cautions, too. They must not turn on the space drive until the ship had warmed for a certain length of time (which the menslator interpreted to Dusty as a vague quantity of minutes. To be safe, Dusty would wait twice that long) and then Scyth would lapse again.

But as the drive went on, Scyth's periods of lucidity waned. His moments of babbling dropped too; and between them both came longer and longer periods of dead silence and heavy breathing.

Yet by the time Dusty drove his car underneath one tailfin, he had a fair idea of how to run the spacecraft.

VIII

Dusty carried Scyth to the salon and dropped him on a divan. He left Barbara to take care of the Marandanian while he went aloft into the control room to take over.

Once inside the room Dusty stopped short.

He was a Hottentot in a powerhouse, a savage in a Plutonium refining plant, a tone-deaf idiot standing before a four-console organ. There were meters and switches and levers and toggles, neatly mounted on gleaming black panels and clearly lettered in shining white. He stared at a pilot lamp labeled :æ:*ææ;æ*æ and wondered foolishly whether the gleam of red meant that the spaceport was still open or whether it signaled that smoking was forbidden for the time being.



He was a Hottentot in a power house, a savage in a Plutonium refining plant.

And Dusty was supposed to drive this.

Stunned, Dusty dropped into the pilot's chair and looked around him in a completely dazed manner. Below his feet were pedals and just below the surface of the slanting panel were a pair of knee-flappers that could be pressed without losing the thrust on a foot pedal. The desk-thing was studded with large levers mounted in curve-segments all carefully marked in the calibrations of the Marandanian language. To his left was a panel filled with push-buttons from the floor to the level above his head where his long arm could reach without standing up. To his right was a similar panel. Dead

ahead was a flat plate that looked like frosted glass and seemed to Dusty about as useful. It neither glowed, nor showed a spot of color other than the very logical reticule-lines which were to be used for aiming the ship. Above the plate of glass was a line of meters and another line of them below.

Dusty shivered. No matter in which way he reached he could touch buttons, or thumb levers or turn dials.

Doubtless the competent Marandanian pilot played this console like a pianist—strictly from practise. A mere matter of training; when the concert master calls for 'A' the musician automatically reaches for the right position and drops his forefinger.

This was no instrument to play by ear.

Or—was it?

"Barb!"

"Yes, Dusty?"

"Barb, find that damned menslator and bring it up here. It might—"

A moment later she came up the stairs with the small instrument in her hands. She gasped as she saw the array of controls and asked, "I thought he said it was easy?"

"To him," growled Dusty. He fitted the menslator on his shoulder by its strap and fiddled with the controls. He hit one setting that made Barbara cry out inexplicably (which irritated him) and then he found another setting that made him feel like a hundred and seventy pounds of toothache (then he forgave Barbara) and after some more fiddling with the tuning and the gain Dusty hit the right setting.

Everything became clear to him.

Directly in front of him was a meter that read "Rhenic Doubler Current" and to one side was a lever labelled "Phanoband Isolator" and some push-buttons marked "Polylateral Overload Reset" and "Primary Exchange Test." The rest, too, were very logical but equally meaningless. "Drive Pulse Synchronizer" must have some definite function because it was a large lever almost in the middle of the desk-panel and what one did with it was undoubtedly taught in the first grade of spaceman's school.

There was a large and interesting handwheel labelled "Drive Angle Trim" which Dusty gathered to be the gizmo used to equalize the drivers so the ship wouldn't yaw in flight, but he was not quite sure. There was another called the "Pre-flight Check Sequence" which probably checked the multitudinous functions of the instruments as it was turned from position to position, but what it did or what it told the pilot made no never-mind to Dusty Britton of The Space Patrol.

There was one that he recognized instantly. It said, reading from left to right "Off, Warm-up, Stand-by, Operate." It was a big four-position hand-lever and it was a good idea, excepting what did Dusty do next?

"Can Scyth help?" pleaded Dusty.

"He's out cold like a Northern Light. Lost blood and—"

"But how'm I to run this godawful thing?"

"I don't know," said Barbara doubtfully. "Try something."

"What?" he asked.

She pointed to a small button high on the front panel beside the glazed plate. It said, "SC/WBN-3 Phanoband 22".

Dusty looked at the nameplate and the menslator helped him translate the nameplate into "Space, Commercial/Non-adjustable, High-power, Emergency—Model Three. Phanoband Twenty-Two."

Dusty looked at Barbara and shrugged. This was an emergency, so Dusty put out a forefinger and pressed the button.

A pilot lamp winked from blue to red and a meter on the forepanel rose. There was a momentary whirring from far below in the big star ship and then along the bottom of the ground-glass looking window in front of him, a small circle began to grow luminous. A man's face appeared.

He was obviously in some sort of uniform; it had that air. The collar was high and the effect was uncomfortable. A pair of gold diagrams glistened on one shoulder. The man looked human enough to be the local desk-sergeant in

costume dress. As soon as the little circle was completely clear he said tersely:

"Distress Call received. Identify yourself, state your position, define your danger, and estimate the time remaining in which you have a factor of safety."

Dusty blinked and then looked at Barbara. She shrugged. Dusty shrugged back and said, "Are you Marandis?"

"This is Marandis Emergency. Identify yourself, state your pos—"

"Stop talking like a robot—or are you a robot?"

"I am not! What is the meaning of this? Using a distress-call band for—"

"This is a distress call," snapped Dusty. "And part of the distress is that I can't identify myself because I don't know the language."

"You'll have—"

"The other part of the distress is that the man who knows all about this is likely to die of a bad accident if he is not given medical attention. So now you know, tell me what to do next."

"Who are you?"

"I am Dusty Britton, if that means anything."

"I don't know you."

"Of course not. I've never been to Marandis. I'm not a Marandanian, just a character of the race your play-mates term 'Backward,' and/or 'Primitive.' But you better do something fast."

"What is the name of the injured party?"

"Scyth Radnor."

"Then your identity is Exploration License K-221-Y. I know Radnor. I must get you off the distress band. Please switch to Space Communications, Band Forty-Five. I—"

"Wait," said Dusty quickly. "As a member of another solar culture you must be aware of the fact that I am not familiar with your equipment. Which knob do I twist and how far?"

The Marandanian gave Dusty instructions and waited for a second small circle to appear beside the first, with a different face in it. This face was older and not in uniform. The man said, "Please explain the nature of your difficulty. I am Gant Nerley."

As well as he could, under the circumstances, Dusty explained his predicament.

"I see," said Gant Nerley thoughtfully. "This is a rather complex problem to solve. Can you state your location?"

"Hardly."

"I suppose not. If we don't know where you are from here, the chance that a non-galactic culture would know where we are from there is indeed remote."

"Haven't you a filed plan of operations?" demanded Dusty, using a tone of voice that indicated that he thought that any culture above the level of the ape wouldn't let people go galloping all over the galaxy, tearing up stars and ruining scenery without first having filed a program and had such program approved by twenty-seven signatures.

"There is a filed plan," said Nerley defensively. "But naturally it is sealed as a matter of protection for the company."

"And no provision for emergency?"

"Only by the consent of the licensed company."

"Then you'd better call a conference at once. Scyth isn't going to last long enough for you to comb the galaxy for us."

"That's why it might be better to let the barytrine field run to completion."

Dusty's voice grew hard. "I wish you birds would stop tossing off a thousand years of our life with the flick of a finger," he said.

"What difference does it make? You'd not notice it, and—"

"Who says so?" snapped Dusty, his irritation mounting.

"Time is of importance only when its passage can be measured in reference to outside events. You have no contact with outside events. Therefore it makes no difference whether you come in contact with us now or a thousand years from now, so long as the same people of your culture are involved."

"Now see here—"

"Permit me to present an example. If the barytrine field went on at this instant, one thousand years from now my successor would pick up the thread of the conversation from the recording we are making, and take on from here. As far as you are concerned the only difference would be a sudden flick of the viewscreen and a rather abrupt change in the facial characteristics of your conferee." Gant Nerley waited a moment to let the point sink in. "Now, since you and I have very little in common, it should make little difference to you whether you spoke to me or to someone else. And as far as I am concerned, I feel the same. I have long since ceased feeling regretful that I cannot retain friendship with the hundreds of thousands of people with whom I must converse. I have almost stopped being regretful of the fact that there are so many worlds that no single lifetime would permit a visit to more than a fraction. I suggest that you try to take a more lasting attitude. You sound as though the troubles of a world you never saw were of prime importance to you."

"Look," said Dusty testily, "A lot of what you claim may be true. But we have a couple of thousand years of observational data on the planets and the nearby stars. You may take a thousand years out of our lives in the twinkle of a second, but then we spend another five hundred on top of that finding out where we are."

"You have time."

"We have not!" roared Dusty. "Move us to a new system and I'll tell you what'll happen. Before we can make a move into space we have to chart the new system completely, because we admit that our reaction motors are not efficient enough to take off without a well precharted course. We must know the orbits of the planets to a fine degree before we dare. Then, before we can make a try for the stars, we've got to spend years and years in observation before we can chart the nearest stars and observe whether or not they might have planets, our astronomy will be put back. Now—"

"Pardon me, but the information I have regarding your system is before me. Your space travel is primitive and any form of real commerce is as yet impossible. This I get from the license application for barytrine operations. Now, how can you justify your statements about interstellar travel?"

Dusty Britton, no matter what else, was a good actor any time he could sit in with a large Virginia Ham to carve. Dusty would never play Hamlet or Julius Caesar; a custard pie in the face was closer to Dusty's art than John Barrymore. This fact provided for Dusty a rather interesting background for the present argument. A student of science could not have faced Gant Nerley without paying deference to the Marandanian's obviously superior knowledge, position and experience. The learned man makes no flat-footed statements; this leads to the odd belief that most learned men are not entirely sure of themselves. It is the bird who is ignorant of all the myriad things that he does not know that can afford to stand up on his hind feet and reel off chapter and verse as though there could be no rebuttal.

So Dusty Britton, who could portray a reasonably convincing role of a wounded hero while mentally contemplating how long it would be before the first preprandial martini, plus being the flamboyant type who never lets a few facts stop his flow of words, was not abashed to let on that he knew a lot more than the Marandanian suspected. Furthermore, Dusty felt that he had Gant Nerley on the defensive, and if he could put the Marandanian off balance long enough to accomplish something, Dusty did not care if Nerley accused him of being a four-flusher at some later date.

Keeping this in mind, Dusty braced himself with little effort and tried to reduce to bafflegab what he recalled of Scyth Radnor's previous statements.

"Interstellar travel is, of course, based upon obvious errors in the theoretical mathematics of general relativity," said Dusty, as though he were reciting some of the science-double-talk usually included in Dusty Britton And The Space Patrol. "Of the many schools of thought which have their own theories on how to explain these obvious errors, the group-velocity field seems to be the most successful. But all of them are seeking some evidence to support their theories, and a couple of them, namely the gravitic and the magnetic-field proponents claim that such evidence has already supported their claim.

Now, if such is the case, you know it will not be long before some practical experiment will disprove the illogic of providing a finite limit to an infinite system. Once this has been established it seems obvious that star-travel is the next step."

"Hmmm—I see. This is a situation that must be considered more carefully. May I ask, Dusty Britton, what is your position in your society?"

"I am Dusty Britton of The Space Patrol," said Dusty with the proper tone of respect. "Commander in Chief of the Junior Division."

"Indeed! A real Space Patrol!"

Dusty nodded at the viewscreen. "It may be a bit ambitious," he remarked with even more deference, carefully studied. "But we feel that there is small point in using a conservative name and then having to change it every couple of years."

"Quite a sensible attitude."

Dusty nodded again. "Fact is," he said deprecatingly, "we would probably be quite a bit more advanced in our space operations if our sister planets were not so inimical to human life. As it is, our extra-planetary operations are limited and will be limited until we can provide the necessary conversions to terrestrial conditions."

Gant Nerley nodded back. "Man is not an adaptable animal," he observed. "He does not change himself to suit his environment; he changes his environment to suit himself."

"That's what I mean."

"Then why do you object so much to this barytrine field?" asked Gant Nerley. "We can always pick you a stellar group less inimical to human life and thus advance you faster."

Dusty grunted under his breath. He had talked too much. "Buster," he said angrily, "logic like that will only get you a fat lip."

Gant Nerley blinked. "Tell me, Dusty, was Scyth Radnor hurt in some altercation over this beacon?"

By this time Dusty figured that he might as well let Gant Nerley have it cold and hard. It would show Gant that the mighty Marandanian was no more distant from the lusty chimpanzee than the terrestrial.

"No," he said flatly, "Scyth was plugged for monkeying around another man's woman."

Gant said, "Deplorable," in a tone of voice that indicated an amused disgust, but not easily identified as to whether over the act itself or the business of being caught at it. "What happened?"

"The other guy shot first," said Dusty, feeling that this was no time to point out that it was he that pulled the trigger.

"I'm not surprised. Most primitives are inclined to be both hot-headed and impulsive."

"Tell me," asked Dusty in a cooing voice, "did Scyth confine his amours to primitives, or is it the custom among Marandanians to consider your mate unattractive unless she can prove it by bedding down with an impressive list of lovers?"

"I don't understand," replied Gant Nerley stiffly.

"Against primitives I can understand Scyth carrying a weapon to his assignation, for protection against the irate cuckold. Tell me, Gant Nerley, has your emotional balance become so stable that you can take a more scholarly view of promiscuity? Or," added Dusty sharply, "do you have big black headlines about triangle slayings and love-nest scandals just like the rest of humanity?"

"Well, now, we—"

"Then don't blame us primitive souls for slugging a guy that's caught off base!" snapped Dusty. "Now, what are we going to do about Scyth?"

"Regardless of his depredations against propriety, he must be given medical attention."

"This I will go along with. How shall we start? I can always take him to one of our hospitals."

"No. No! You must not."

"Why not? We're quite competent on gunshot wounds. We're probably more used to them than you are, as primitives with impulse and hot blood."

"Please. Let's not be facetious over any man's misfortune."

"In blunt words, the life of a character caught in an awkward situation is more important than someone else losing their familiar stellar scenery and a couple of thousand years of climb up from the swamp of ignorance?"

"That is another question which I'm sure we can solve. Now—"

"Look," said Dusty firmly, "you agree to take measures for our safety and we'll agree to take measures for Scyth's. Do you understand exactly what I mean or shall I explain in very blunt words?"

"That is blackmail."

"It's worse than that. But we're primitive, and therefore lacking in refinement. As far as I am concerned, Transgalactic can keep their secret of our position locked in their sealed file. Scyth can die, and Bren and Chat can spend the rest of their lives marooned on Mercury."

"No. That wouldn't be right. You must bring Scyth back home."

"That's a fine idea! May I suggest that your ship is not as familiar as mine?" Dusty did not mention that the only control room he was familiar with was the one on the Gramer Production Lot, which was an aggregation of fantastic levers and flashing lights and futuristic three-phase busbars which had a most profound effect upon the imagination of the youth of the land but no effect upon space whatsoever.

"This can be taken care of. As a spaceman, you can understand the principles. They are simple. You can follow directions for flight."

"Yes? And which way do I go from here?"

"Not so fast. First, Dusty Britton, tell me the present condition of Scyth Radnor."

"Wait."

Dusty went below. Scyth was in a state of shock. His temperature "taken with the flat of Dusty's hand" was chill—and there was a film of perspiration wetting Scyth's body. The breathing was shallow and the face was pale. Scyth's pulse was weak and the heartbeat thin.

Dusty turned a light blanket over the Marandanian and then went back to report.

Gant Nerley said, "In the salon you will find a medicine cabinet. The instructions are simple, any intelligent being with a menslator should be able to follow them concisely. How is the bleeding?"

"Stopped. Clotted by now."

"Take care of Scyth, Dusty Britton. We'll figure out something for you."

"How about this barytrine field that's running away with itself?"

"We'll stop it. Behind you on the auxiliary panel you will see a knob and a pilot lamp, probably orange colored. Turn the knob to the left."

Dusty did, and the lamp went out.

"That's it. I see that Scyth has the usual sloppy habits of his kind. No label. According to space regulations the operator is supposed to slip a label into the frame above the auxiliary control whenever he has anything extra set up. I'll mark that oversight down on Scyth Radnor's record. Now—"

"What about Chat and Bren and that variable-star maker?"

Gant Nerley grunted. "If they're not keeping a close eye on the barytrine field detector, so they can shut off their own equipment when it fails, I'll revoke their licenses! They must be looking at the temporal field, or at least keeping watch."

"We hope."

Gant nodded thoughtfully. "Now," he said, "this being an emergency, I'll open their course-plan so that I can direct you through space. Don't turn off the viewpanel, Dusty. I'll be back in a few minutes."

As soon as Gant Nerley's face disappeared from the viewpanel, Dusty turned to face Barbara. She was standing far to one side, out of range of the viewpanel, and stifling a giggle. She let it bubble through her fingers as soon as Dusty caught her eye.

"Funny as hell," he said. "Me—I'm hysterical."

Barbara sobered immediately. "Honest, Dusty. I wasn't laughing at you. I was laughing with you."

"Why?" he demanded sharply.

"Because you really fooled that bird. Dusty Britton of The Space Patrol. Yes, I can navigate a ship."

"I'm going to. Want out?"

"I wouldn't miss this for the world. Glad we've got the whole galaxy for you to make mistakes in."

"Stop making fun," he snapped. "Let's try and think of something sensible, Barb. Too bad we haven't time to take a run back to the city."

"What good would that do?"

"Well, you could show 'em that bauble you're wearing and I could try the menslator out on 'em, and maybe between us we could convince 'em that there's something more in this tale of mine than wind."

"That's an idea, but it's out."

"I know. But—"

"Dusty, you'll have to carry it to Gant Nerley yourself."

"Carry what?"

Barbara shook her head impatiently. "Think!" she cried. "Dusty, this license might be rescinded if we can show that Sol has evolved above the minimum level of acceptability."

"Yes?"

"Then go in there with your head up and let 'em know how we're built."

Dusty waved at the field of instruments on the control position. "Open my yap and let 'em know how ignorant we are? We should have a couple of

scientists along."

Barbara shook her head. "No," she said slowly. "One of the marks of a real scientist is that he usually considers that he knows a lot less than he does. You're better off. You don't know enough to confuse yourself. Besides, Dusty, you're an actor."

"Um—er—Jeeks! Hang on a mo' will you? I've an idea."

Dusty loped down the stairs to his car and opened the compartment behind the front seat. It was his emergency kit; it held his Dusty Britton uniform, the complete regalia of The Space Patrol complete with Dusty Britton 'Blaster' concealed against the days when Dusty found himself trapped in public and could not appear out of character.

He changed in the car and went back to the control room.

Barbara took one look at him and nodded slowly. "You're a gaudy sight," she said. "But maybe that's what it takes."

Dusty slapped the 'Blaster' at his hip. "I look authentic enough except for this hunk of hardware," he said. "Hell, it isn't even as useful as a dress sword."

"Your revolver? Oh—still on my living room floor."

Dusty unbelted the holster. "I shouldn't have to go armed everywhere, should I?"

"I suppose not."

"All right, then. How do I look?"

Barbara smiled thinly, "Dusty, no one on earth would ever accuse you of being anything but a Hollywood actor in that get-up. But a man from halfway across the Galaxy itself might not know about these things. You might be an Admiral of the Swiss Navy. You're impressive-looking. Just don't get pompous."

"Just you remember that I'm Dusty Britton of The Space Patrol and don't giggle when I start dishing it out."

"I won't. After all, I call myself an actress, you know." She looked nervously at the viewpanel.

"Are you all right?" he demanded.

"Yes. I'm nervous but I'll be all right."

Dusty went over to her and put his hands on her shoulders. "Take a deep breath," he commanded. She did. "Now let it out slowly." She did that, too. "Now," he said softly, slipping an arm around her and leading her to the stairway, "You come down below and relax. Pull yourself together, Barb. We'll make it—somehow."

"Got any ideas?"

"Not yet. But—"

Above, the voice of Gant Nerley came back. Dusty raced aloft and apologized for having been absent. Gant was nodding with admiration at something below the level of the view panel, probably something on the desk.

Gant looked up after a moment and said, "Dusty Britton, this is really a remarkable route. Truly fantastic. So well hidden, and yet right within our grasp all of these centuries! Well, you shall see, Dusty. And doubtless you will agree."

"Okay," said Dusty, "let's get going."

"Not so fast, young man. I'm waiting for the direction-finding stations to report so that I can determine where along this prospected route you lie."

"We're about two-thirds of the way out from the center, I believe," offered Dusty.

"That's a rather inaccurate generality. You know where you are and we know where we are, but we must know where we are with respect to one another before we can make contact. Now—" Gant's voice stopped suddenly as something caught his eye above the lens of the viewpanel, and he looked over Dusty's head, apparently, so intently that Dusty himself turned to see what Gant was staring at. He saw only instruments, and realized that Gant was looking at another panel-section above the one that communicated with Dusty's panel.

"Um," said Gant. "You would appear to lie in what we call 'Sector G-18, Co-ordinate 307, Galactic Angle 215.86-plus degrees, South altitude-angle 1.017-minus degrees, Co-frame 9654.' Now, Dusty, in your terms, where lies the Galactic Center?"

Dusty laughed. The tone of his laugh was half bitter and half a note of self-disparagement. "Sorry, Gant. We frame our reference from Terra, naturally."

Dusty breathed a sigh of relief at having boned up on enough science to play his part convincingly.

"I do not quite understand what you mean," returned Gant.

"We compute stellar positions in latitude from the angle above or below the equator of Terra, which we call 'Declination' and in longitude by their rise as the planet rotates, which we call 'Right Ascension'. Therefore the so-called 'Celestial equator' is a projection of the Earth's equator upon the sky, and the colures pass from celestial pole to celestial pole, which are projections of Terra's axis. Now, since the Earth's equator is tilted with respect to the Earth's orbit, and the Earth's orbit is tilted with respect to the Galactic Equator, I'll be darned if I know how to explain in mutual terms. Oh, we assume that the galactic center is in a region of the sky we call 'Sagittarius' but that is meaningless."

"I agree. Wait a moment."

Gant turned from the window in Dusty's viewpanel and walked away from it by several yards. He worked over a complicated keyboard for some minutes and then returned.

"Dusty," he said, "I think we can handle this as follows. To your left hand near the top of the control board you will find a key-lever marked Phanobeacon. Pull it towards you."

Dusty looked, found the key, and pulled. A bright spot of light appeared on the view panel, high in the left hand corner. "That is the true position of Marandis," said Gant Nerley. "If you tried to make it at transgalactic speeds you'd plough into about forty stars and hit about nineteen gas-clouds. You'd either blow up, or spend the rest of your life running at safe velocities.

However, if you take off and steer your spacecraft so as to put that beacon spot on the calibration lines G-705, F-318, you should find the next rift-beacon somewhere near to the crosshairs of the viewpanel. Got it?"

"I think so."

"Good. Now, for take-off instructions. Ready?"

"Ready."

Gant Nerley began a running patter of instructions. Those favored few who have ever seen the control room of a spacecraft can possibly grasp the implications of the problem. One does not step into the pilot's chair of a complex device such as a galactic cruiser, push a pedal and then steer any more than a Wall Street Accountant could step into the cockpit of a six-engine airliner and take off, just like that. There was the pre-flight checkoff, probably performed by the competent Marandanian Pilot in a matter of minutes, and quite possibly done with an automatic reflex action which would permit the accomplished pilot to daydream about the girl on the next planet meanwhile; only the appearance of the wrong pilot-lamp response would bring him out of his automatic response with an abrupt recognition of something awry.

But Dusty was not a pilot, and certainly not a pilot of a Marandanian Spacecraft. So the pre-flight checkoff took almost an hour. Nearly ninety-nine percent of the time Dusty was following Gant Nerley's instructions blindly: Is the pilot lamp registering power source showing red or green? Is the spacelock indicator showing closed? Turn the atmosphere control to Internal. Set the autogravity corrector to Controlled. Co-stator circuits to Regulated; antimagnetic response dial to zero; space-coordinate servo control to Stellar Display. Planetary Drive to Automatic Threshold; match the Gravitic Constant to the Power Delivery. Set the Master Control to Pre-flight Warm-up.

"Now," said Gant Nerley, "take it slow and easy. Take the 'Tee' bar gently. Find the thumb-buttons and press them both evenly; spread your knees against the paddles under the control panel slowly and press the Force pedal with your right foot. Tell me, what is your trans-atmospheric velocity?"

"It says 416."

"Too high. Press the Compensator pedal with your left foot until the TAV meter reads 312."

"Now."

"Hold it that way until the Matter Per Cubic Meter indicator drops below the red line."

"The TAV meter is dropping below 312."

"Good. Let up on the Compensator pedal and depress the Force pedal more. Keep the TAV meter at 312."

"The Matter Per Cubic Meter indicator is below the red line, Gant."

"Free the Compensator pedal. Push the Force pedal all the way home and kick it to the right. Now read the Trans-atmospheric velocity meter."

"Dropping rapidly."

"Good. And the MCPM?"

"Dropping rapidly."

"Excellent. Spread the knee-paddles wide and lock them. Have you a reading yet on the Space Velocity Meter?"

"Just getting off the peg."

"Um—it is a little early. But that's all right. It will arrive in due time. Keep an eye on the Foreign Body Indicator, Dusty. Any reading?"

"No."

"Good. Don't touch the 'Tee' bar, Dusty. That's the steering mechanism and it is in neutral. Is there any indication on the viewpanel yet?"

"Not yet."

"Haven't enough velocity yet," said Gant. "But when it appears, it will look like a star map. Now, the central cross-hair is the point of aim of your spacecraft. If the star you want lies, say, to the upper left, move the 'Tee' bar forward and to your left. That will swing the ship in that direction and you can line up the drive with the target. Also, since angular position is important when moving in three free dimensions, twisting the crossbar of the 'Tee' will

cause the ship to rotate on its axis. The map will turn in the direction, apparently, but it is really the ship turning. That is—"

"I'm beginning to get a presentation now," said Dusty.

"Good. Dim and reddish, isn't it?"

"Yes."

"Fine. Now get this straight and clear: The phanobeacon is the control beacon for direction of angular curve. In other words, it takes three points to define the orientation of a plane in space. These three points are you, the star-beacon or course-marker which you will find directly, and the main terminal-beacon which is the phanobeacon. You must drive your ship in the proper plane when making a curve or making any turn. Follow?"

"Yes," replied Dusty, trying to think it out. He was far from certain about all this, wondering why it was all necessary. He went over the instructions in his mind, made no more sense out of it than the first time, and then decided to accept it without trying to figure out the reasons. After all, Gant Nerley and his folks ought to know what they were doing.

"Now," said Gant, after a moment, "In order to orient yourself, you must line up the Phanobeacon on the point of aim. Take the 'Tee' bar firmly, one hand on either side of the axle. Find the thumb-buttons on the handle. Press them all the way in and lock them home with a slight sidewise pressure towards the center. Got that? Now, lift the 'Tee' bar straight up until it is high enough to manipulate with ease. Be careful, don't move it sidewise!"

The last admonition was wasted. Dusty lifted the 'Tee' bar gingerly and not too evenly. The stars on the viewpanel danced dizzily, swiveled, and flowed across the plate. The bright phanobeacon spot moved from the plate along the bottom, danced back in view on a brief curve, and left again along a flat slant. The 'Tee' bar clicked into place and the stars stopped dancing with a snap. Dusty moved the 'Tee' bar gently and the stars flowed upward until the phanobeacon re-appeared.

"Got it," he said shakily. He moved the 'Tee' bar very gently until the phanobeacon was centered on the screen. Or, rather, almost centered. It

moved in jerky little circles like the sights of a rifle in the hands of a tyro.

"Fine. You're doing well with strange equipment. Now, on the panel you will find a switch marked 'Co-ordinates.' It will be set on 'Rectangular' and you must flip it to 'Polar'."

The switch changed the cross-hair pattern of the viewpanel from the horizontal and vertical calibrations to a circular pattern with only the main center hairlines remaining. Angle-lines radiated out from the center, crossing the circles.

"Now, Dusty, inspect the radius-line marked G-705. All the way around. Do you see a winking star?"

"No."

"Um. I was hoping we could do it the easy way. The sealed course-plan is not too clear, for which I don't blame Transgalactic. All right. We'll have to do it the hard way. Move the phanobeacon down until it is almost on the lower edge of the viewpanel. Now flip the 'Co-ordinates' switch to the left, leaving it in the bottom position marked 'Polar.' You'll find that the toggle has an 'H' type pattern of motion, laid flat-wise."

The polar co-ordinates disappeared completely from the center of the viewpanel and centered around the phanobeacon spot. They made larger and larger arcs as the circles approached the top of the panel.

"Now this is going to be tricky. You must twist the 'Tee' bar slowly and let the ship rotate, but you must also move it so that the phanobeacon stays near its present off-center position. But before you do this, let me explain what you are actually doing in space. Picture a needle-shaped spacecraft with a line along the axis running out before the ship, marking the line of drive, or direction. At some distance from the line lies a spot which denotes the phanobeacon. Somewhere out beyond, there is another spot that must be sighted within the confines of an angle not greater than the angle made between the point of aim, or line of drive, and the imaginary line running from the nose of the ship to the phanobeacon. So you must cause the ship to rotate on a false axis, making the line of flight describe a cone of revolution with the phanobeacon on the axis of the cone. Now, go ahead and try."

"Okay." Dusty moved the 'Tee' bar and the stars moved in jaggledy little scallops along a greater arc. The center of the beacon held the polar lines,

but they moved with the stars and with the beacon. It made Dusty dizzy and his eyes began to ache. "What am I looking for?" he asked plaintively.

"Look along the outer circles for a winking st—"

"Got it!"

"Good. Turn the 'Tee' bar to neutral," said Gant. "Return the 'Co-ordinate' switch back to the center of the 'H' pattern. Center the stellar course beacon on the point of aim."

The winking star flashed at Dusty like a flag. It danced crazily as he manipulated the 'Tee' bar with all of the thumb-handedness of the rookie pilot on his first attempt at the controls. There was so much to do, so many things to handle, so many motions to make. Dusty gripped the 'Tee' bar tightly, too tightly. When he let go with one hand to flip a switch or to make an adjustment, the grip of his other hand moved the bar. It became sweaty and sticky, then it became slippery and he gripped it even tighter, which made it worse because his fine control left him as he strove to hold the handles tighter and tighter.

In a jagged line like the trail of a rising smoke, the winking star proceeded to the center of the viewpanel. There it hung, wobbling around in tiny circles and occasionally making a brief jerky dart to one side or the other. Dusty mopped his face and the beacon star jumped; he grabbed the handle again and the star leaped across the center and wobbled on the other side of zero-zero.

"Got it," he said in a quavering voice.

"Now rotate the ship until the phanobeacon is on the vertical hairline. Then flip the switch to 'Rectangular' again."

The stars scalloped around in the viewpanel until the phanobeacon was on the vertical line. The field leaped a bit as Dusty found the 'Co-ordinates' switch and returned the calibration-presentation to the horizontal and vertical hairlines.

"Now?" he asked.

"You have a bit of time. Be certain that the star-marker lies firm and true. Be careful!"

Dusty gripped the handles and tried to steady his shaking hands. Then, because he had no more complexity of motions to make, he relaxed a bit. The dancing star-field slowed its mad vibration, which calmed Dusty's jumping nerves still more.

He leaned back in the pilot's chair slowly, his grip on the 'Tee' bar lightening and becoming more true. He looked at the beacon star and knew what Chat, Bren, and Scyth were working toward.

It lay there on the center of his panel like a winking flashlight. Lost in the star-field, which showed a myriad of points, some dim cloudy stuff, and a band of milky white, the beacon would have been nothing without that steady wink ... wink ... wink. He, himself, was lost. He had not the foggiest notion of where he was, excepting that Mother Terra must be far behind. Sol, a smallish, yellowish, completely average dwarf would show nothing to call attention to itself from the distance of a few light-years. Yes, somewhere back behind him lay Sol and his planets. But the winking beacon on Dusty's viewpanel was like a banner waved from a distant shore.

No man is alone so long as a lighthouse flashes its message of safety, or warns against danger.

Dusty took a deep breath. "Barb!" he called.

She came up the ladder. "Call me?"

"How's Scyth?" he asked.

"He's doing all right. How're you doing?"

Dusty nodded boyishly. "Look, Maw I'm flyin'," he told her with a chuckle. "Martin Gramer should see me now. This is simple like a duck's ear, and I —"

Barbara screamed and Dusty whipped his head back to look along the direction of her horrified eyes. To the viewpanel.

One of the stars, lost in the glitter of the distant background had detached itself from the immobile sky. It was moving, forward, and its glow was brightening. It came hurtling towards them like a white hot cannonball. One

second it was no more than any other star, distant, aloof, and cold. Then it had exploded into a disc that expanded like a released puff of gas. It came toward them like a ball of fire hurled into their faces.

Dusty yelped and twisted on the 'Tee' bar and the stars rolled dizzily across the plate—but not until the white hot monster had flipped past in a quick wave of heat and a final flare of light which made a small section in the back of Dusty's mind recall the effect of having a foil-filled flashbulb fired during a still photography session.

Shaking, Dusty's grip on the 'Tee' bar tightened and he moved the lever in tight little jerks until the stars returned to the proper positions and the Phanobeacon was properly centered.

Gant's face showed concern. "What happened, Dusty?"

Dusty told the Marandanian, and Gant smiled knowingly. "Don't worry about it. It will happen again and again, and maybe worse. But so long as you keep the course beacon centered properly, you will pass by—and not through—those interfering stars. Now, as soon as your beacon star shows a disc, steer up to keep the beacon centered on Line H-001. Once you pass the beacon, look for another beacon on Line F-312 and bring the point-of-drive to center on the new one. Follow?"

Dusty nodded at Gant's image on the screen along the bottom of the viewpanel. Another star detached itself from the backdrop of stars and hurled itself into Dusty's teeth. The actor flinched but held his drive. The star passed in a bright flash and a quick wave of heat and was gone. Dusty licked dry lips and forced the grip of his hands to relax. Far to one side another star passed in a majestic sweep, too distant to bring them either heat or more light than the ones called 'fixed' on the viewpanel.

Dusty eyed the star-beacon suspiciously. Was it showing a disc yet? And how much time did he have to shift the drive once the disc became certain? Dusty felt a cold wave wriggle down his spine and he knew that cold beads of sweat were beginning to ooze out of his face; he was remembering the staggering speed with which the first star had come leaping at him.

Another star passed him in its characteristic wave of light and heat, and Dusty realized that what looked dangerously close on the viewpanel was in reality quite distant. It meant that so long as his ship was pointed into a clear space, there would be no danger of running into a star no matter how precarious it looked.

But the cold sweat came because the beacon star lay winking at him dead in the intersection of the crosshairs that marked the drive.

Disc? Did it show a disc? Does Sirius show more of a disc than Polaris?

Dusty's hands pulled the 'Tee' bar slightly to move the winking eye ever so subtly upward. That way he would not be aiming his spacecraft dead into the searing hot maw of a variable star. He took a shaky breath and relaxed.

Gant Nerley shook his head. "I see what you are doing, Dusty, and you must not. You'll make a wide curve and get off the beam. Or worse, you'll hit a star lying close to the course. You have no idea of how wide you'll run. Center it up, Dusty, and keep a close watch, for it will become a disc. You'll have time. Relax."

Reluctantly Dusty returned the 'Tee' bar to the central position, and the star winked through the crosshairs at him, itself no larger in diameter than the width of one line. It was not obscured by the lines because of the construction of the panel, a design that Dusty could not quite understand. Dark lines should have hidden the stars behind them, but on this gadget they did not. He looked closer and found that the stars themselves lay on top of the lines rather than under them, and he wondered how they managed that stunt. It was, of course, a matter of design. Dusty's experience had been with small telescopes, but this device was not an optical device, so the simple laws of optics did not obtain. As he watched, the winking star became a winking disc and Dusty's nerves twitched.

When had the change started? Dusty realized that he had been half-hypnotized by the wink ... wink ... wink that meant both safety and ultimate danger. The disc was expanding rapidly, and as Dusty tried to move the disc to Line H-001, the edge of the winking beacon expanded faster than the point of aim moved. He wrenched the 'Tee' bar hard and saw the crosshairs move sluggishly below the exploding circle. Then the beacon flashed past in a wave of heat far greater than any of the other stars, and he was blinded by

the light for a second or more. But as the blindness died, there on Line F-312 there was a distant wink ... wink ... wink.

X

Dusty gripped the 'Tee' bar and started to turn the ship toward the new beacon. His approach to dead center was ragged—he overshot and over-corrected, but finally he made it. And then with a burst of good sense, Dusty released the 'Tee' bar very gently and leaned back in his pilot's chair. The crosshairs stayed on their winking beacon.

Gant Nerley nodded. "Turn the presentation to 'Polar' again, and keep a sharp eye out for a slow beacon along Radius Q-103. You probably made a wide curve around that other beacon and you may be a bit too close to a gas field. You'd burn up in milliseconds if you hit it at your present speed. By the way, what color is the presentation now?"

"It's getting lighter. Sort of yellowish-white, like."

"Good. But if and when it begins to blue-up a bit you'd better let up on the 'Force' pedal by a notch or more. Competent pilots can run with their screen in the violet, but you're far from being a competent pilot." He saw the look on Dusty's face and added hastily, "I mean that you've had no experience in galactic travel, Dusty Britton. You're doing magnificently so far. We'd best take no dangerous chances, though, until you have driven interstellar craft as many hours as you've driven your own interplanetary ships."

Barbara made a choked sound and then covered it by saying, "I see the slow beacon, Dusty. Out there on Circle D-212, along Radius Q-103."

It was pulsing slowly, rising to full brilliancy over a period of more than a minute and falling again, never really winking out to invisibility. It lay alone in the star-field; the gas cloud behind it must be of the same nature as any of the so-called 'dark nebulae' or dust clouds that obscure the stars behind it. But it was far to one side (Circle D-212) and it seemed reasonable to view it calmly.

"How much time have I?" he asked Gant Nerley.

"About fifteen minutes."

"Good. I want a cigarette and a drink."

It was with increased confidence that Dusty swooped around the next beacon and headed on towards the next—and the next—and then around a long curveway limned by four of the winking beacons and once more along a long field-free course towards a winker that lay dead ahead for quite some distance.

There was one quick jog between two beacons set at an angle like the flags of a slalom run on skis; a wide 'S' curve around the outside of the first, up and over, between, then out and around the second beacon in a long ogee to locate the freeway to the next beacon star. There was a quick turn that took the plane-locating phanobeacon off the screen for several minutes and then another one that put the phanobeacon almost on the crosshairs, and then another slight turn that put the phanobeacon on the lower corner of the viewpanel again. It was, according to Gant Nerley, a "most remarkable rift!" At which Dusty shrugged because he had never seen any other rift. It looked plenty complicated to Dusty, and he shuddered to think of what a really tortuous galactic passage would be like.

They passed by a vast luminous cloud that lay on the spacecraft's beam for minutes. It looked like a matter of mere miles that separated them from it. It was marked by two of the slow-winking beacons, as if that were necessary. The luminous cloud reminded them of a lake, seen from an automobile driving along a highway. They could not see the inner star that provided the energy for the luminosity of the cloud, and eventually they left the luminous cloud behind them. They zipped between the elements of a star cluster that drove at them with multiple waves of heat, and on and on they went with Dusty Britton driving his Marandanian spacecraft like a child running a motorboat, following instructions shouted by a careful, protecting parent.

This did not make of Dusty Britton a space pilot any more than turning the valve on a radiator makes one a heating engineer, or replacing a light socket makes one an electrician. But Dusty began to glory in it; his confidence grew high as his skill increased.

His touch upon the 'Tee' bar became light and sure of itself. He no longer waggled widely or jerked the bar when a deviation became noticed, Dusty corrected his course with deft touches like the driver of an automobile. He was learning, and filled with a self-confidence he had no right to feel, but did not know enough to be scared about. Dusty Britton, who had never been in a space rocket in his life, drove a galactic spacecraft across the galaxy under what can be called "Dual Flight Training."

Which was all right, so long as the trainee has enough space to make mistakes in. Dusty literally had galactic reaches and these were well marked against the pitfalls. And if Gant Nerley's face radiated confidence and his voice sounded cheerful it was due to Gant Nerley's knowledge that constant admonition, warning, and cries of horror would only cause more trouble than Dusty Britton's meandering course.

But flight is easy, whereas landing is the most difficult maneuver in the universe.

So by the time Dusty Britton was homing on the main phanobeacon of Marandis itself, Gant Nerley had his plans. Dusty Britton was not going to barrel that spacecraft down tailfins first like a screaming elevator that might come to Velocity Zero at a plus or minus a half mile from Ground Zero and maybe a plus or minus thirty seconds of Drive Turnoff; to drop like a plummet or ram the spaceport with a planet-shaking crash or burn a crater in the 'port with full drive still warping the space below the ship's tailfin.

Dusty Britton came to a full zero-zero-zero landing a million miles above Marandis. He came to a grinding halt high above the planet, looked around dazedly, and asked Gant: "What makes?"

"Keep your drive at one gravity thrust," said Gant. "Stand by for Pilot!"

The last order was delivered in a ringing voice as though it were a standard procedure.

To Dusty, familiar with the tactics used by seagoing liners upon entering port, standing by for a pilot was quite a sensible practise. If the Captain of *The North America* permitted a pilot to bend the big liner along Ambrose Channel, through The Narrows and into New York Harbor, Dusty Britton saw no objection to having a pilot come aboard to bend the big spacecraft

down past whatever dangers might be presented by moons, meteors and cosmic junk.

And Gant Nerley, not knowing how Dusty felt about spacecraft piloting, hoped that this procedure sounded like Standard Operating Practise.

Dusty replied in ringing tone, "Standing By for Pilot!"

Gant took a deep breath.

Minutes later a small scooter hauled alongside and a Marandanian pilot came aboard and took over. He smiled at Dusty and said, "I'm Nort Wilgas, Pilot."

"Glad to have you aboard," smiled Dusty. It all sounded very familiar; The Space Patrol had borrowed liberally from the clichés of naval procedure and courtesy and he had been through these lines at least once in every picture. "I'm Dusty Britton." Then he remembered the role he was trying to play and added, "Of The Terran Space Patrol."

"Have a good passage?" asked Nort Wilgas.

"Yes. A bit tiring. After all, I've never driven a galactic spacecraft before. Frankly, I'm about flat on my face."

The Marandanian pilot looked into Dusty's face with a perplexed frown. "You know," he said, "It's just occurred to me—you drove this thing all the way on duty!"

"Yes."

"Twenty-three hours!"

Dusty suddenly felt tired. He had been too busy with the board to think of it before. He had been running on nervous energy, but now it had about run out. Dusty had been this way before; so long as there was something that had to be done he had done it, but once the need was over, he invariably came unglued and slept the clock around.

"Yes," he said. "I had to."

"Man! What stamina!"

Dusty yawned and came unglued on the divan opposite the one that Scyth Radnor occupied. Nort Wilgas nodded at him and then turned to Barbara.

"You can relax too. I'll take over."

Dusty Britton was fast asleep when the spacecraft made its landing on Marandis.

XI

Dusty awoke to find the sunshine streaming in through a small porthole and lighting the cabin cheerfully. The smell of fresh air was in his lungs, a pungent, pleasant smell faintly of cinnamon or nutmeg but not quite either. He recalled that he had folded out on the divan in the salon, now he was in one of the cabins below the salon level. He wondered how he had arrived.

He stretched his muscles, the cool sheets felt pleasant against his back. Then he wondered who had undressed him and how anybody had been gentle enough to do the job without waking him. He looked around the cabin expectantly and as he looked, his door opened and a woman came in.

She was wearing white from cap to slippers and Dusty pegged her for a nurse at once. She was wholesome enough, but neither her face nor her figure would have stopped any traffic on Fifth Avenue. She carried a book with a finger slipped between pages to mark her place and in her other hand she held the Marandanian equivalent of a cigarette. A pleasant curl of smoke rose from the lighted end.

"Hello," she offered brightly. "And how do we feel this morning?"

"We feel fine," grunted Dusty. "And we'll feel better after a shower, a shave, and some of that smoke you're using. I'd also enjoy a change of clothing."

"We took the liberty of having your uniform cleaned. The shower and shaving gear is in the bath—there—and as for the cigarette, I can offer you one right now."

"Give," said Dusty with a grin. She handed him a case and snapped a lighter for him. He puffed and found that the stuff, while far from tobacco, was tasty enough. He took a deep puff and let the smoke filter out through his nose.

The nurse said, "I hope you don't resent sleeping in the—ah—"

"The raw? I do it all the time." Dusty took another puff and swung his feet overboard onto the deck. It was not deliberate, Dusty was just uninhibited and the question of wandering across a cabin dressed in nothing did not even occur to him. The nurse said, "I'll be waiting for you in the salon." She left, not precipitately, but with a certain air that removed all embarrassment.

Dusty showered and shaved and dressed in his cleaned uniform. When he got to the salon, Barbara was there already, also freshened and cleaned.

"So this," she said, "is Marandis."

The nurse nodded cheerfully. "This is Marandis. You'll have to tell me how your Terra is; I've never been anywhere near that far from home, you know."

"Sure," nodded Dusty. "But now that we're on Marandis, what do we do next?"

"Oh. I'm to escort you to a formal meeting of the Bureau where you'll meet Gant Nerley face to face."

"How's Scyth Radnor?"

"Why, he's doing very well. He's hospitalized and he'll be out and howling for the skin of the man that shot him in about a week."

"He'd better take a month off for practise, first," grinned Dusty.

"Or," chuckled the nurse, "leave other men's women alone."

"Yes," agreed Barbara.

The nurse nodded. "You're very attractive," she said with no trace of jealousy or envy. "I can see Scyth getting side-tracked along your direction. I am a little disappointed in Scyth—seems to me he could do better than a frauland for you."

"Better than a what?"

"Frauland. That bauble he gave you. You wouldn't know, of course, but it comes from Selira, a stellar colony not far from here. It's incredibly cheap."

Barbara tore the chain getting the bauble away from her. "Next time," she promised sharply, "I'll plug Scyth Radnor myself!"

The nurse shuddered a bit. Dusty merely laughed and said, "So now we know where we stand. And now knowing, I'm hungry."

"Of course. We'll all dine at the meeting."

"Oh?"

"Naturally. You're here, aren't you? Marandis is not going to send you back without a chance for you to present your case. There is a joint meeting of the Bureaus of Galactic Navigation and New Colonial Affairs. It will start with a formal breakfast during which no business will be conducted. Then, once you are all acquainted with one another, the business of the day will be discussed and a decision rendered."

She led them to the spacelock and Dusty Britton had his first glimpse of a Marandian spaceport. There was precious little to see, which made it even more stunning to the senses.

The size of the place was completely obscured by spacecraft which stood like the trunks in a pine forest. Most of the craft were larger than Dusty's and so obscured his vision. Between those nearby (which were rather wide-set) there were others at a little distance, and beyond them there were still others, and behind those others were more and more and more until all that could be seen were the tips of the upthrust noses. The horizon was an irregular pattern of pointed shapes that was somewhat reminiscent of the Greek egg and dart moulding of ancient architecture.

Through some of the more distant lines of sight, the far spacecraft had a haze around it, as though it were miles and miles away.

There was not a building to be seen, only spacecraft.

Dusty gave up trying to penetrate the forest to the edge of the 'port and directed his attention to his nearby surroundings.

A road wound around in a zigzag manner, meeting and dividing around each ship. There was an empty landing block not far from them, and after a bit of puzzled interest—the shape of the block caught Dusty's memory—he decided that the landing block was hexagonal. So were all the rest of the landing blocks. Hexagonal pattern, like the well-known hexagon tile floor;

the road was the marker-lines between the hex-shaped landing blocks. Those that were empty showed the effect of heavy masses parked on them; a bit of char now and then; a chip or a crack, probably made by a rough landing; a deep seam repaired with some sort of cement or concrete (or whatever the Marandians had devised or discovered as a superior material) and at least one place where the edge of the block had been chipped deeply as though the pilot had missed his landing point and come down on the edge of the hexagonal block.

As they looked, a muted whirl attracted their attention and they turned to see a ship lowering itself out of the sky to come down in a slowing vertical drop that ended at the edge of a curtain of nearby spacecraft. The landing ship inserted itself in the pattern behind ships until only its nose was visible. Then to one side—and apparently with no warning, a ship nosed upward, gaining speed rapidly until it disappeared in the bright blue sky above.

The nurse said, "We land a ship every thirty seconds. There's a take-off every thirty seconds, too."

"That is a lot of activity," said Dusty, swallowing the daily figure with some amazement—7,200 ships landing—a like number taking off—every hour, night and day. The traffic added up to a rather monumental figure. No wonder they required a huge spaceport.

"Marandis is the center of Galactic culture," said the nurse proudly. "And this is only the spaceport that handles affairs of the Space Administration Department. Each of the many Departments of Galactic Government has its own spaceport. The one at the Department of Space Commerce is the largest because that is the one that takes care of incoming transports carrying the necessities of living."

"Don't you do anything for yourself?"

"We have no room. Marandis is an urban planet. The only parts that are not built-up are preserves, parks and recreation-forestry. There is nothing on the entire planet that does not serve directly toward Galactic Administration, in one manner or another."

Dusty nodded. He could grasp this even though the magnitude was great. By simple proportion, if it took one complete city to administer the government for a country, it should take one planet to administer the government of a galaxy. He wondered even then how they managed to get it all in.

He smiled and made a wave at the landing ramp. He had seen everything he could see from the little platform outside of the spacelock.

At the bottom, in the zigzag road, was a lone, low-slung vehicle with a man in a simple uniform leaning indolently against the wheel. He was smoking a cigarette which he tossed onto the landing block as they came down. He fired up the thing under the nose of the car after they were inside, and as soon as the door slammed, he let the clutch out with a rap and the car jack-rabbited into motion. They took off from a standing start like a frightened deer at about five degrees lift so that by the time Dusty and Barbara had pulled their heads forward from the jerked-back angle, the car was about thirty feet in the air and arrowing forward above the road. The speed climbed rapidly until Dusty estimated something near to a hundred miles per hour.

The driver was, of course, cutting the tips of the corners between the hexagonal blocks in a die-true line of flight and naturally paying no attention to the zigzag road below them. Since the spacecraft were all standing in the center of their particular blocks, like a bunch of chessmen parked on a tile floor, there was plenty of space between the ships themselves for such passage. Even at their thirty-foot altitude, which raised them to a point on most ships where the bowed-out flanks were quite wide, there was room to spare.

And now that they were in one of the aisles, distant buildings could be seen dead ahead. It must have been ten miles from their landing block to the edge of the spaceport.

The driver barreled along this aisle with the self-assurance of any taxi-driver, hooting his horn now and then as they came to what seemed to be a major intersection of the zigzag road below. Dusty wondered worriedly what happened when two of these characters met in a draw, because the man seemed to pay no attention to any other noise but his own, which he made with great confidence, in the other guy.

Dusty was beginning to wonder about the need for any road below when his question was answered by a caravan of heavy trucks making their way along

the road. They zipped over the caravan and were gone by the time Dusty realized that air-travel was not for heavy cargo.

The buildings at the end of the aisle between the spacecraft loomed larger. The driver whipped along at his thirty-foot altitude, making no attempt to climb over the buildings which were growing taller and more massive at a frightening rate. Dusty's palms went wet; the buildings had seemed minute when they turned into the aisle, but now they were tall and massive and millions and millions of windows could be seen, with magnificent arches between the buildings spanning the gaps.

The aircab whipped across an empty perimeter about the hexagonal-pattern of landing blocks, sped above a low building, and howled into the tiny space between two buildings with an arch above and a roof below, and then went into a flat climb. The car rose slowly in the canyon between the buildings that lined the street below. There were people working in those buildings, men and women that sat at their desks behind windows and paid no attention to the passage of a hundred-mile-per-hour skycab within forty feet of them.

Then the car was above the roof-level but it kept to the street-lanes. Below them were the streets, and in the valley was slow-moving traffic, ground cars and air-cars that ran at different levels to avoid intersection-collisions. Up in the higher strata were the fast-moving aircabs, each moving in its lane, and each lane for a different direction. Even with separate lanes the traffic was a turmoil; constant jockeying to gain position, to avoid trouble, to move a level higher or a level lower so that a corner could be turned without entering the intersection at the wrong level.

To make a right turn the driver jockeyed himself to the top of the altitude allowed that line of traffic, and in the block before his turn he rose above his lane, made his turn, and then entered the right-bound traffic pattern from below, mingling with the speeding aircabs. To make a left turn, the driver dropped to the floor of his lane, fell below, made his turn, and mingled with the left-bound turmoil from above their upper limit of altitude.

They raced along in the middle-altitude at high speed; cars above them, below them, to the left and right, before and behind.

"My God!" breathed Dusty, "New York at rush hour—in three dimensions."

Their driver turned and winked at them. He flicked a lighter with one hand and lit the smoke that was hanging from one corner of his mouth. "Yeah man," he drawled. "Some of them guys should ought to take lessons."

Then he turned back to his job with a shrug, lost a hundred feet of altitude in three hundred feet of run, and whizzed around a corner and fitted his aircab into a space between traffic that was just large enough to let him in without scratching paint. The other cars moved up, aside, down or sped or slowed to give him elbow room. He fought them for position, dropping on a descending run through this cross traffic until he whipped out of traffic on a spiral over the roof-top of one of the buildings.

Here the driver phlegmatically put the aircab into a tight corkscrew that dropped them onto the roof. Dusty got out slowly, testing the stiffness of his knees after the ride. He helped Barbara out next and the nurse came out on the other side at the same time.

Then they were almost roofed as the aircab took off on a flat, screaming 'U' turn that lofted him no more than ten feet, whipped across the street between levels and swooped him down on the opposite side, where he hit the other roof without a bounce and came to a fast braking stop beside a man who had flagged him.

The man got in and the aircab whiffled off the roof in a crazy climbing turn and burrowed into the fast traffic lane above. It forced its way into the mass of traffic and was lost in a matter of seconds.

"Holy Rockets!"

Barbara wiped her damp forehead with the back of a shaking hand. "Oh—for a film of this!"

Dusty grinned weakly. "Shucks, Barb. What's a fender for if you don't fend with it?"

Quietly their nurse turned from the spectacle and led them to a roof kiosk and down some steps into an elevator....

The operator cut the ropes and let them drop slightly slower than the free-fall constant of the planet Marandis, leaving their stomachs somewhere up on the hundred and ninety-first floor. He braked the elevator somewhere down below-below-below, and their innards caught up with them in such a sudden rush it buckled their knees.

Along a magnificent corridor and through massive carved doors opened for them by men in uniform, and then they were ushered into a vast ornamented room with a vaulted ceiling, tapestried walls, and a polished floor. Deep armchairs were waiting around a huge table that glistened with polished metal and blinding white cloth, the severity broken by color of dish and fruit and fluid. Soft stringed music filled the air that was also lightly scented.

As they entered, the music bridged from the stringed fugue to a magnificent orchestration and the scent changed subtly from languid sweetness to a pungent aroma that compelled the senses to pleasant attention. The soft-key lighting swirled across the vaulted ceiling and changed into a colored brilliance that made the blood leap high.

The music slid into a soft passage and a vibrant voice announced:

"Dusty Britton, Commander in Chief of The Junior Division of The Terran Space Patrol. Barbara Crandall, Thespian and Vocal Musician of Terra. In attendance, Lela Brandis, Mistress of Extra-Marandanian Medicine."

The music crashed, the scent came heavy and sharp, and the lights flashed like the licking of summer lightning and came to rest outlining them brilliantly.

Gant Nerley crossed the huge room and held out his hand to Dusty Britton.

"We need no introduction, Dusty Britton," he said in a ringing tone. "I say 'Greeting' to you with all my heart!"

Another stab of music, a touch of cinnamon-scent, and a play of lights.

Gant Nerley turned. "Stop the dramatics," he commanded. "What are we, children to be impressed by theatrical tricks?"

The music shifted back to the string ensemble, the scent smoothed out to something pleasant and pungent, and the lights faded back to their neutral medium-key. Dusty thought that if this lights-and-music stuff was strictly off the cuff, ad-lib, someone was a past master at the art of extemporaneous

composition. He liked it. And if it took Marandanian children to appreciate it, you could give him ten years in school and call him the Marandanian child.

Gant Nerley was holding out an elbow to Barbara. She took it and the Marandanian led her towards the head of the table. Dusty looked around; then he offered his own elbow to the nurse—Mistress of Extra-Marandanian Medicine, Lela Brandis.

It was many years before Dusty identified the things he had for breakfast. It was exotic and well-prepared; none of it was remotely familiar but all of it was good.

Then over the after-dinner drinks and smokes, Gant Nerley rose, rapped the table with his knuckles, and proposed the problem for the day.

"What are we going to do about Sol?" asked Gant Nerley seriously. Dusty eyed the Marandanian soberly. "Leave it alone, I hope."

"You realize what you are asking?"

"My God! Do we have to go through all that mishmash again?"

"Again?"

Dusty slammed the table with his fist hard enough to make the glassware jump. "Again and again. I'm getting sick and tired of explaining all the many reasons why none of us want to move to another star and lose a thousand years. And then being told that after all it won't hurt a bit, and besides this move is necessary—and if we don't move willingly we'll be moved anyway forcibly."

"Why are you so angry?"

Dusty looked at Gant Nerley and sat down wearily. "Because," he said patiently, "all of us know that no matter what, you're going to go on and do it anyway—but not until you've forced yourself to believe that you have convinced us that we should accept this kick in the pants gracefully."

"It isn't that simple."

"No?"

"No, it isn't. You see, we are bound by our own laws to hold to certain programs under certain conditions. It is the conditions which prevail that we are attempting to define, outline, determine, and pin down so that we know what lawful action may be taken."

"You sound like a bureaucrat explaining away an awkward situation. Just what do you mean by conditions and programs?"

Gant picked them off on his fingers. "There are the following," he said. "First would be a race—remember I am talking about all the races of mankind strewn across the galaxy; the races that are us, you and we and all the rest that stem from a single source, the origin of which is lost in the antiquity of a hundred thousand years. So, first would be a race which was still in the growing-up stage, say the mound building, early agricultural, perhaps later, in early metal. An age of no true scientific grasp; what little of science they know has come by guesswork, blundering discovery and hit-or-miss experiment. Such a race could be moved across space without a qualm, because it would only bring about a rather deep period of superstitious horror and a religious fear. A few hundred years later the tale would be completely discounted, because the astronomers would be rising and stating flatly that no agency in the universe could change the constant stars. The old sky would be wiped out of men's memory in a couple of generations, although it might remain in myth and fairy tale for a long time. Such a set of conditions would permit the moving program without a qualm."

Gant looked at Dusty. "Understand?"

"Sure," replied Dusty indifferently. "Go on."

"Then on the other end of the scale we have the advanced race. They have discovered the phanobands, know about space flight and perhaps have colonized the planets of other stars say within ten to fifty light-years. A race of this stage of development would understand and grasp the problem quickly. Then having been shown the problem, they would make the move willingly and no doubt help, because they would understand that their destiny is a part of the Galactic Destiny."

"Oh, yeah? You mean to say that if Marandis were found to lie across the road like a stone wall you would all happily toss Marandis into a barytrine

field for a thousand years?"

Gant smiled serenely at his objection. "Well, doubt it as you will, but we would. Of course, we know that no such case would ever come up. But if it did—"

"Y'know what you remind me of," snapped Dusty. "You remind me of a parent explaining to his kid that this castor oil is good for the kid, and that if the parent needed it he would take it with a happy smile—excepting of course that the parent does not need anything of that nature. We have an old adage: he dies well who never faced a sword! I think it applies here. Well, go on, Gant. Tell me where Terra lies in your scale of values."

"That's what we are trying to determine. You are obviously not of the pre-aware stage. You have your limited space travel and your historical records which will preclude any attempt at forcing the affair upon you and causing you to put the change as superstition that would be wiped out."

"Thanks."

"On the other hand you are not of the advanced stage which could accept a change in your night sky without trouble, nor will you accept it willingly."

"How true. Now this brings us to the impasse, doesn't it?"

"Yes."

From across the table a man waved for attention. "It's more than that. The moment Dusty Britton opened the distress phanoband, the secret of the galactic rift was let out. Like everybody else, we put direction finding equipment on the signal and have it located rather well. Then we went back through our files and found that as far as we can tell, Sol was mentioned as a possible beacon by one of our early exploratory parties. One that disappeared. One that—"

"So what?" barked a man down the table from Dusty. "Seems to me that Intercluster sits on its duff and waits for us to find rifts for them."

"Transgalactic isn't the only outfit with a spacecraft," snarled the man from Intercluster angrily. "We've done our share."

"Not on my books," said the Transgalactic representative.

Intercluster eyed Transgalactic sourly. "What's the matter?" asked Intercluster softly, "Are you mad because Intercluster happens to have records on the rift you re-discovered?"

"Re-discovered my—"

Intercluster turned to Gant. "I leave it up to you," he said. "Our records show that we, too, have rights to this rift."

Transgalactic hammered on the table. "Like hell!" he roared. "If you have rights, why aren't you using them?"

"Because you and your gang concealed them from us until Scyth Radnor got into trouble. A fine bunch of incompetents you are! A fine group to be representatives of our culture. You can't even keep your hands off native females—"

Barbara rose with a single lithe motion and hurled the contents of her glass in the Intercluster man's face. He staggered back, floundered back into his chair, landed hard and tilted it back on hind legs to go over backward in a crash.

"Native female?" spat Barbara.

The room went breathlessly silent; the music stopped on a flubbed note; the scent soured in a brief wave as though the man at the valves had miscued. The lights flickered awkwardly.

Barbara stood there tense and ready. Her breasts were pushed against the nylosheer of her dress; her stomach was flat and hard. She was poised like a healthy wild animal daring any onlooker to try to tame her.

Dusty rose lazily and pushed her back into her chair with a hand on her shoulder. No other man in the room would have dared to lay a hand on her except Dusty. This he somehow realized, and it gave him personal gratification to know that he had once more done that which the Marandianians would not have dared.

Then he went over and picked up the Intercluster man with one hand, standing the man on his feet like a puppet.

"We apologize for reacting to your unfortunate choice of words," he said smoothly. "We admit to being a bit primitive and impulsive. I came unarmed," and he pointed to the band across his hips where the Dusty Britton Blaster belt had protected the whipcord from the sun, "because we are advanced enough to realize that we are impulsive and perhaps a trifle inclined to act before considering the matter fully."

He turned away from the man and sauntered over to Gant Nerley. "I apologize again," he said. "But I do suggest that our nerves are a bit short. After all it is hard to sit here and listen to your friends and fellow-citizens discuss the ways and means of making use of that rift through the galaxy without once recognizing that we poor devils have to move out whether we like it or not."

Gant smiled nervously. "I am trying to appreciate your position," he said. "And in a way I do. But you must try to appreciate ours. We are not taking anything away from you that you will miss. After all, Dusty, what do you stand to lose, really?"

Dusty swallowed. It dawned on him what he was doing and why. And also how he had managed to get away with it so far.

And in these fractions of a second, Dusty probably matured more than he had grown during the great part of his life.

He realized suddenly that he was only Dusty Britton of The Space Patrol and as phony as The Space Patrol itself. To date he had done as good a job of wool-pulling as the best statesmen or scientists, but only because he was an actor. He had succeeded in convincing the whole bunch of them that the cultural level of Sol was higher than it was. A scientist would have admitted his lack because that was the way scientists operate. A businessman would have been baffled, and a statesman would have tried to cover his indecision in a gout of flowery language that would be known for what it was by this bunch of high-brain characters.

But Dusty was an actor, blunt and not too smart. Modesty is not part of an actor, while the ability to submerge himself is. He had become Dusty Britton of The Space Patrol and the hero of a hundred adventures in space among a

people who were hard and fast because they were still in struggle against their environment. He was tall and strong and young and handsome, and he was Dusty Britton, fast on the draw, hard on the trail, and the bes' dam' cabba-yero in all Mehi-co and he had them all convinced that he and his friends spent their time racing around in dangerous, imperfect spacecraft powered by reaction motors.

He was Dusty Britton who had plugged Scyth Radnor for playing games with his woman. Then Dusty Britton had taken the controls of a completely foreign spacecraft and had driven the ship halfway across the galaxy through danger and God-knew-what (Dusty did not) horrors and possible fates. The fact that Gant Nerley and a corps of engineers and a bank of computing machines had supervised Dusty's every motion and move did not detract from the feat in their eyes. It added, because of the sheer guts of a man who would enter an alien ship and have the self-confidence to touch the tiniest push-button.

He sauntered over to Gant Nerley and said, "Well?"

Gant Nerley was impressed with Dusty's swagger and self-confidence. So were the rest of the men in the room, with the exception of the representatives of the two shipping companies, and they had chips on their shoulders. So Gant Nerley looked around from face to face and then said, in an official tone:

"It would appear that Terra is of a level of development that mitigates against immediate action. Therefore we shall declare a recess, during which time we shall study the Terran people. If Terra measures up, other steps must be taken."

There was a chorus of "Aye!" and the sound of chairs being pushed back and the noise of feet on the floor. The babble of voices arose as the members broke into little groups and began discussing the problem.

But Dusty did not hear them. The self-confidence had oozed out of him and he slumped in his chair, staring at some shine on a bit of the table silver, trying to think of something other than the horrible certainty. For while Dusty Britton had bluffed the Marandianians, he knew without a shadow of a doubt that his bluff was being called and it would not stand up. All it would take was the Marandian Investigating Committee scouring Terra to find one single man who had one shred of reason to believe that matter could exceed the velocity of light. Oh, there were such people. But the man who professed such opinions believed it because he wanted to believe it; because he hoped someday that it might be accomplished. He was the man who shrugged off experiments that followed the rules and acted according to the equations. He was the man who had faith but no proof.

Beyond a doubt, the report of any such committee would recommend that Terra be bundled into its barytrine field with no delay, and that Sol be nudged into the three-day variable needed for the beacon on this particular dogleg of the journey across the galaxy.

Dusty had succeeded in his own way, but now he knew that it was not enough. He, himself, had convinced them that Terra was worthy of notice. The rest of Terra would let him down. Still lost in his own unhappy thoughts, he became vaguely aware that the babble of discussion was stopping and that one man was raising his voice to get an audience.

It was the Transgalactic representative. He was standing by his place at the table, talking in the tone of voice used by a professional lecturer hammering home an unpleasant fact:

"—obvious by the animal ferocity of this Terran, his threats and his willingness to plunge into physical combat, that he and his kind cannot be of high culture. I am asked whether or not we may judge an entire race of people by one man, and I agree that we cannot. But then view the reaction of his companion who flares up in a fit of red, raw anger, taking offense at being properly catalogued. I ask you, gentlemen, is there any excuse for this? Am I not a native male of Marandis? Is she not a native female of Terra?

"And so by their actions, both violent in nature and unpredictable in direction, they have shown themselves to be uncouth. Who knows what offense they will take next? Does a man among us dare to speak freely with either man or woman of Terra alone and unprotected? No, because no one can ever know beforehand what peculiarity of their own limited semantics will be rubbed the wrong way, setting them into a violent fury. Dusty Britton has boasted that he can take any of us out and wipe up the street with us. This cannot be denied. But what does it prove? Only that his shoulders are broad and his back strong and his fists hard. And that he has been trained in violence.

"Now, gentlemen, consider this next argument: What has Terra to lose? No more than a familiar night sky, really. The time under the barytrine field will pass without their notice. As for the time lost in respect to the rest of the galaxy, since they have had no contact with it, they cannot be affected by the loss. They prate about losing a thousand years of advancement. Consider how soon they would be taking to space if we had not found them. Might it not be yet a thousand years before contact with the galaxy took place? Yet as it stands now, this man and this woman will live to see galactic commerce, whereas they would be dead and gone without ever knowing of the galaxy if Marandis had not found them. And having been granted that, they still show the ignorant rebellion of children.

"They have not the foresight to understand that so far as they are concerned, less than a week of their apparent time will pass before the ships and men of Marandis will land on Terra in its new surroundings, to treat with them, to lead them, to educate them, to bring to Terra all of the glories and benefits of galactic civilization—no, gentlemen, *to return to Terra its galactic heritage, lost so long ago. Its birthright returned!*

"And yet what response do we get? Objection and rebellion and threats of violence. So I ask you, are we to be frightened by this small primitive world that lies like a barrier across our path? Are we to be cowed by a show of force? Are we? And if we are, shall we run in fear from a race of men who bear missile-propelling weapons?

"Look at Dusty Britton and his companion. They sit there angry, possibly planning their own form of revenge to take place if we have the temerity to proceed. Then let me ask you, supposing they do object? Suppose they do resent our meddling in their small lives? Are we to be frightened of bomb and gun—we who can put them back into their barytrine field and keep them there until they are willing to agree? *And without the loss of a life?* Gentlemen, this whole meeting reminds me far too much of parents who try to argue logically with children over bedtime instead of packing the infant off. Who knows what is best? Child or parent?"

The man from Transgalactic paused a moment to let this point sink in. Then he said, "Gant Nerley, I object to your proposal. We need no more investigation. We know what these Terrans are and how they react. They offer little to Marandis at present. They are no more than a responsibility to us and as such they owe us our superior rights. Therefore, unless I am ordered at this moment to cease and desist, I am going to proceed. Do I hear such an order?"

A babble of voices rose.

"Gentlemen," said Transgalactic, suavely, "I offer you a short and quick route to the Spiral Cluster."

He stood there for fully a minute listening to the clamor of individual discussions going on in the smaller groups around the table. Then he hit the table with his fist, bowed sardonically to Dusty and Barbara, and strode out.

Dusty looked at Gant. "Can't we do something about this? Can that guy go do as he pleases?"

Gant shrugged. "We are a government that guides but does not rule, suggests but does not demand, recommends but goes not force. I can and

will put a stop to his activity providing that you show direct evidence that Terra and Sol are of importance in their present location, that Terra has something to offer Marandis, that you are not what he claims. However, if what he said is true, then what he is about to do is acceptable."

"But we—" and Dusty stopped short. He had no argument strong enough to convince this Marandian that Terra would lose anything but its own jealous prestige.

Dusty stood up slowly. "Come on, Barbara, let's go home. At least we can be among friends. I'd hate to be marooned here while Terra was smothered in the barytrine field."

Barbara stood up and leaned against his side. "Yes, Dusty," she said in a throaty contralto.

Gant smiled wanly. "I'll see that you get home," he said. "Forgive us, Dusty. You'll really lose little and gain much. I—"

Dusty looked at Gant. Then he looked down at Barbara. Then up at Gant again.

"So I've failed," he said in a low voice. "I've tried and failed. And I am aware of the fact that Terra will not lose much. That isn't the point. It's just that I, Dusty Britton, am a personal failure. I should like to be able to say that I don't give a damn what other people think, but I can't. I care a lot what other people think, because for the next forty or fifty years or more I've got a living to make, and making a living is a lot easier if the entire world is not convinced that I am a no-goodnik. But then, who am I to stand in the way of galactic progress."

"Dusty, I regret that the rest of your people will not be able to see the thing I am going to show you. Maybe you can describe it when you return. Come with me."

Gant led them from the hall, then to a moving walk that hurled them out and across one of the flamboyant arches between buildings. Here Gant stopped to display his credentials to a man in uniform, and to sign a register that also listed Dusty and Barbara and their home planet Terra.

They went along a corridor that curved gently; through a heavy metal door that opened on response to a signal sequence delivered against a button.

The room inside was vast, truly vast. It was a vertical cylinder and it must have been more than a thousand feet in diameter and three or four hundred feet tall. They stood inside of the door on a narrow metal catwalk that ran completely around the circle, its far side lost in the distance and the dimness, for the room was not lighted from above, but from below.

It was a pleasant glow, a flat, hazy, wispy glow from a gas-like cloud that floated in the room a hundred feet below the catwalk ... a scale model of the galaxy.

It looked like any photographs of one of the galaxies taken through a telescope except that this model was dotted here and there with winking pinpoints and stringed through and through with thin lines that glowed in many colors, some solid colors and some in two-color spirals, like coded wire cable. Here and there were faintly glowing spherical volumes containing many stars, or rectangular volumes confined by planes of faint color-glow. Certain of these clusters were linked with other clusters by the zigzag lines that wound and interwove around and through in a tangled skein.

Gant Nerley picked up a small cylinder from a rack on the railing of the catwalk. A narrow pencil of light pointed out, and he aimed it towards the center, some five hundred feet out to the middle of the hall. "Marandis," he said. Then he brought the pointer-light across towards them slowly, to stop a hundred feet from their position. "Sol," he said. "The lines are courses surveyed and registered by the various companies, you can gather that the colored stars are our inhabited systems and the volumes register certain clusters. That faint greenish-yellow course that ends out there by Sol is the Transgalactic course set up to reach from Marandis to the Spiral Cluster which lies almost at our feet."

The magnificence of the spectacle was enhanced by the silence in the room. The galaxy, it seemed, lay at their feet and with no irreverence, and only awe, the viewer felt as though he were standing by the side of God, looking down at his Work.

In a hushed voice, Dusty asked, "Is this where they survey the courses? Couldn't figure out a way around Sol?"

Gant laughed sympathetically, breaking the hushed awe. "Look at it and think, Dusty Britton."

Dusty looked, and Barbara looked, both in awed silence as Gant Nerley went on:

"In that model, which looks like a wisp of gas, there are fifteen billion individual pinpoints. Think, Dusty. One-five, comma, zero-zero-zero, comma, zero-zero-zero, comma, zero-zero-zero stars in one galaxy. Across the breadth of this room it lies, scaled down to represent the hundred thousand light-years of its diameter at the rate of a hundred light-years to the foot. Eight and one third light years per inch, Dusty Britton, so your Sol and your Sirius lie about an inch apart. Now, Dusty, in order to make the stars visible, they must be above a certain intrinsic size, and in the size of the stars the scale of the model is violated. Each tiny glowing point is about one-sixteenth of an inch in diameter. That makes the scalar size of the stars about a half light year in diameter. The diameter of the colored lines that represent courses is of the same magnitude, so as we go into the model—as we may—we will find that the courses touch, intersect, and lie tangent to stars that are actually far from the flight in real space.

"What I am saying, of course, is only a new concept of something that you already know, but pertaining to another subject with which you have every right to be more familiar. Take a globe of your Terra. It is excellent for locating areas, finding cities, lakes, oceans, mountain ranges; anything gross enough to find physical size upon the map. But you cannot use it for a road map to direct you to the home of a friend, because the details of such a trip are much too fine. So we use it for large-scale mapping, but could not possibly use it for the delicate business of course mapping."

"But if you enlarged a section?"

Gant Nerley nodded. "It has been tried. No good. You see, Dusty, this was made by going deep into space and making stereograms from all angles. The transparencies are used in projectors all around the hall. But as you may know, the finest photogram loses definition when it is enlarged too

much. As for delicate operations, Dusty, just to prove our point we are going to enter the model."

Gant led them to a control panel in the railing and from a sheet of paper in his hand he set the dials.

The vast circular runway lowered all around the hall and the galaxy-model rose, giving the appearance of turning upward past them. "We are coming down toward and below the plane of our galaxy at the scalar rate of about a hundred thousand light-years per minute," said Gant. Then a segment of the catwalk detached from the wall and went forward on a long girder.

The bright pinpoints leaped out at them, giving Dusty the same feeling as he had had in the space flight, except that the model lacked the waves of heat as the little pinpoints passed. He looked at Barbara and watched the tiny points plunge into her skin to disappear, then reappear behind her, as if they passed through her body harmlessly. He looked at his hand as the points streamed through, and he wagged his fingers around a cluster and watched them twinkle. They penetrated clusters and dark-cloud areas, placed where fifty stars occupied a volume of less than a couple of cubic inches, spots where dusky, shapeless masses represented globes of fifty light-years in diameter. Rusty caught on. Thoughtfully he looked at Barbara and made a rough computation that he and she were a couple of hundred light-years apart. His eyes, he thought, must be about thirty light-years apart, and the diameter of his head, at eight-and-a-third light-years per inch—

Dusty began to feel light-headed.

Through and through the model ran the colored lines, tangled and skeined and then they were facing the point where the greenish-yellow course-line ended.

Above the control panel was a faintly glowing sphere about two inches in diameter.

"Sol?" asked Gant.

Dusty shrugged. "Sol? How can we—"

He leaned forward and set his right eye close to the pinpoint of light and looked outward. Was it—could it be—familiar. He changed his angle of sight. Was Galactic North aligned with Terrestrial North? Dusty could not remember. The center of the Galaxy? Somewhere in or near Sagittarius, he believed, but Dusty was not familiar with the constellation. There! Was that the Belt of Orion? It looked strange, distorted. The constellation as he remembered it of old, was not like that. Pinpoints, of course, could not begin to look like these tiny discs, or vice versa. Was it this that made them seem unfamiliar or was it that he was displaced in scalar space by enough light-years to distort the constellation? Was that—there—Polaris and the Dippers, large and small and Andromeda? Or, thought Dusty with wry self-disgust creeping into his mind, was that W-shaped thing Cassiopeia? He wished that he had paid more attention to astronomy.

Pleiades? He shook his head. That was a cluster and unless one remembered very carefully the configuration as it looked from Sol, the conglomeration of stars would probably look about the same from the same number of light-years from the opposite side.

Sol—if that sprinkle of glow were Sol—must be near bright Sirius. An inch away and a double star. And Alpha Centauri should lie about a half inch from Sol and it should be a fine trinary; two bright ones in a binary and a less bright one making the triple. And Procyon—or was that only a single like Sol? He ran through his sorry list of stars remembered as being within fifteen or sixteen light-years of Sol, and was appalled to see the number of pinpoints that were surrounded by that tiny sphere that represented the sixteen light-year diameter. His mental catalogue had holes in the listings—more hole than listing, he considered truthfully.

Confused thoughts and vague remembrances plagued him. Wearily Dusty shook his head. For here, up close to the sprinkles themselves, he knew that they were not scaled. How could the scale show a binary when the size of the stars was scaled at a half light-year in diameter? If that bright one were Sirius as he supposed, it was a single blob because Sirius and its companion were quite lost in the half light-year diameter of the glowing spot that represented the system. And so, of course, was Centauri. No, one could not

scale a hundred-thousand light-years down to a thousand feet and hope to retain enough detail to calculate a course.

He nodded unhappily and looked along the green-yellow line that ended at Sol and realized that at least at one place in the course there was a change of direction that was so shallow that the diameter of the line representing the course was so wide that the ship, in actuality, only traversed space from one side of the line to the other, changed course, and returned to the first side.

Dusty leaned forward again, looking along the yellow-greenish line that marked the Transgalactic course. At the far end he noted the wink ... wink ... wink of the star-beacon, not much different than it had appeared in Scyth Radnor's spacecraft. "Where," he asked, "does their course lead from Sol?"

"The prospectus, of course, is not shown as finished," said Gant. "But we can show it momentarily." He pressed a button and a dotted line of yellow-green flashed into view, extending from the end of the solid line out and out until it was lost to their view through the star-field toward the outward Spiral Cluster.

Dusty looked at the line. "I suppose it isn't to scale or anything," he said. "But I can't help hoping—Gant, look, suppose this model were truly to scale, couldn't they save themselves a beacon here?"

"Save a beacon?"

Dusty nodded and the little speckles blinked at his eyes. Gant shook his head. "This model was built in the hope that we could play gods standing in our galaxy with a measuring stick. We failed because we are no nearer to the stature of gods than this model is to the stature of the galaxy itself. We cannot play gods, Dusty Britton."

"I'm not trying to play God," said Dusty solemnly. "I'm just thinking that if you can move a planet away from a star you want to convert into a three-day variable, you might be able to take your barytrine field and slap it around this star here," Dusty pointed to one with a forefinger, "Then you move it aside and that gives you a long run from this beacon to that beacon—missing Sol by a full inch, or—eight-and-a-third light-years."

Gant blinked. Slowly, he said, "Move the star—" and let his voice trail away into a mutter. "Move the interfering star—" he repeated again. "Then

—my Lord!"

"What's the matter?" asked Dusty.

"Yours is the glimmer of an idea that makes for the birth of a new concept!" breathed Gant. "Take it from there, Dusty. Don't you see? Move a star and straighten out one dogleg, move two and iron out the course even more. Maybe we could drill a free channel completely through from Marandis to the Spiral Cluster. Maybe from Marandis to Star's End, to Vannevarre, to Rescrustes—perhaps from Laranonne to Ultimane across the whole galaxy, a hundred thousand light-years of free flight without a change in course. I —"

A tiny spot of light came crawling along the yellow-green course to disappear into the tiny pinpoint of light that represented Sol.

Gant said, "That must have been Transgalactic, returning to Sol to—" then Gant jumped. "Dusty! Come on! There's no time to waste!"

He hit the buttons on the control panel viciously and the little flying catwalk swung noiselessly back across thousands of light-years of scaled distance to fit into its niche once more. The circular catwalk rose high above the wispy model to its former position.

XIII

Of course Dusty had expected there would be quite a difference between his handling of Marandian spacecraft and the professional. But he did not realize how great this difference was. In a larger ship than Scyth Radnor's, spearheading a conical flight of twelve more ships, he rode behind the pilot and admired the smoothness of the man's operation.

The color of the plate was high in the blue-violet and the stars leaped out of their background to whip past with hardly a flick. Beacons fairly buzzed and they grew into flaming balls and were gone behind as the pilot moved the 'Tee' bar with a deft motion of one hand and used the other hand to flick back and forth across the controls, changing the viewpanel co-ordinates and adjusting the various factors for flight. He skirted gas fields dangerously

close and zipped between the cluster by the double zigzag with a swaying motion, then humped the spacer down tight and made a dead run for it.

And behind him in a cone came the rest, in tight formation, conically arranged below the leader in tiers, three, four, five.

They soared around another beacon, its flashing fire bright blue and the coronal glow reaching out for them, and then the pilot was calling out numbers and a man at the computer was punching keys. On the viewpanel before them lay another beacon, winking ... winking ... winking.

Behind them, a continuous tape was running through the recording machine, playing its words on the phanoband communication channels: "Calling Transgalactic. Government Priority and Emergency! Calling Transgalactic! You are to disable your barytrine generator, you are to discontinue all operations at once! By Order of the Bureau Of Galactic Affairs!"

A man sat tense in his chair peering at a greenish screen that had a halo-spot in the middle. The halo was growing larger, but so slow as to be almost steady. The man held a micrometer thimble between his thumb and forefinger and was turning it slowly, keeping a pair of dark lines tangent to the bright edge of the halo. From time to time he would call out a figure which another man would pluck out on a keyboard.

"Why don't they answer?" breathed Barbara.

Gant smiled sourly. "Because they are going to go through with it if they can."

"But—?"

"They have every legal right to maintain communication silence, even though at the present time there is small point in maintaining secrecy about this rift. Their legal position is one of fair safety; one cannot be convicted of disobeying orders that one does not hear."

Dusty eyed Gant angrily. "You mean to say they can't hear that signal?"

"Of course they hear it. But can you prove that they hear it?"

"On Terra we have a maxim that ignorance of the law is no defense. This is to keep people from shooting people and then claiming that they didn't

know that shooting people was forbidden by law."

"Very sensible. We have the same laws and the same interpretation," smiled Gant. "But in this case we have a different situation. As of the last acknowledged contact with Transgalactic, and specifically that part which is dealing with Sol and Terra, they had every right to proceed. The law has been changed. Now it is up to the law to see that the change in law has been properly delivered to the interested parties and that the change is acknowledged. Follow?"

Dusty nodded. "*Ex post facto* sort of thing. If you pass a law forbidding neckties on Tuesday, you cannot arrest a man for having appeared on Monday without one."

"Right."

"But this is already Tuesday."

"But to be effective, newly-passed laws must be properly posted. Then must be acknowledged from the farthest point in space. And Transgalactic is playing communication-silence."

Dusty grunted angrily. "And that was the character that yelped about our vengeful nature? Isn't he guilty of the same?"

Gant Nerley nodded. "Of course! Aren't we all of the same cut of human?"

The phanoband signal went on:

"Calling Transgalactic! Discontinue all operations by Order of—" and so forth.

The squawk box on the wall said, "Calling Gant Nerley with report."

"Report!"

"Report slight increase in phanoradiation high in the subnuclear region. Cross semi-collateral traces indicating an increase in lower-level nuclear activity."

The squawk box clicked off and Dusty looked with puzzlement at Gant Nerley. "What was all that?" he pleaded.

"He means that Transgalactic is hard at work. The lower level of nuclear reactions has increased in intensity, meaning in simple prediction that the

business of making a variable star out of Sol is under way."

The Marandanian with the filar micrometer on the barytrine detector grumbled. "It's going to be a bit rough."

"Why?" asked the pilot. "If it weren't for that barytrine we'd never find Sol out of that mess dead ahead. We'd be canvassing the stellar region around there for weeks if we didn't have a focal point—"

"I know," grunted the detector operator. "First you need a barytrine field large enough to make a homing run on, but then once you're home you'll want a tiny one so you can locate the generator precisely. Well, you can't have 'em both, Jann."

Jann Wilkor shook his head. "I wish I'd made this run before. I could make it faster."

Gant pointed at the screen and nudged Dusty. The color-scale was still high in the blue-violet and there were a couple of places on the viewpanel that were a dead black, tiny spots that did not move as Jann Wilkor's delicate touch corrected the course. Spots burned out of the substance of the panel like over-exposed film burned through.

"It takes a master pilot to make a run this fast. Even so, we're taking a rather high risk. But if the channel was free and open from Marandis to Spiral Cluster, with only a big phanobeacon at either end, we could make it with the screen burning black-violet. We may even have to develop a new supraradiant material for ultra-high velocities."

"How fast can you go?"

Jann Wilkor soared around a beacon and centered on the next before the flicking wave of heat was gone. He did it easily and with the negligent reflex of the master pilot. "Fitt Mazorn took one of the high speed jobs into intergalactic space for a speed run a year ago and claims to have made it from Laranonne to Ultimane in slightly less than an hour. Or," corrected the pilot, "an equivalent distance, out in deep-deep space. Some of this is probably guff; I doubt that he could do it. That's a hundred thousand light-years per hour and just a bit fantastic. Trouble is that the phanobands propagate at a finite speed, according to Hahn Tratter, and therefore the true velocity is difficult to check, since no one has been able to measure phanoband velocity."

"At any rate, it's fast," said Dusty, who did not understand half of what the pilot said.

Gant nodded. "It's fast. It's what we'll be doing in your clear channels, Dusty. That will make you rich and famous, that idea of yours."

"Ifing and providing we can get there in time."

"No matter. If Terra is lost to you, you'll still—"

"Look," said Dusty, "if that bunch wins out, I'll—"

"And I won't blame you," replied Gant.

There came a double report. The man on the barytrine detector said, "Barytrine field just went into the second phase," at the same time that the pilot said, "Last lap!" and turned his point of aim around the beacon to center the hairs on a small star that did not wink.

"Our next problem is to scour Terra inch by inch to find their barytrine generator," said Gant worriedly.

Dusty groaned. He thought of the trackless wastes of the planet; the Upper Amazon jungles, the tundra of Alaska and Siberia, the hidden reaches of Africa, high Tibet, and for that matter the cornfields of Iowa and the wheat fields of Saskatchewan. The fathomless, staggering area of the sea bottoms was too vast a hopeless search-problem to contemplate.

Gant looked at Dusty. "It's bad, Dusty. I'll not fool you, but it's bad. We have perhaps a day or two, perhaps three. We're late. By the time we arrive, the phase-two growth will be heavy enough to cause leakage-reaction in our detector and render the detector completely ambiguous."

"Meaning what?"

"What I said. That we must scour Terra inch by inch. And here is where you must help."

"Me?"

"Yes. You must issue orders to your Space Patrol to comb the landscape. You must find that barytrine generator."

Dusty looked at Gant Nerley blankly. "You realize what you're asking? That within a matter of hours we must set up a land-scouring search and completely cover the entire earth? I haven't even got the foggiest notion of how many million square miles of earth there are, let alone the ocean-bottom which we couldn't even touch, lacking the equipment."

"They wouldn't plant it on a sea bottom."

"No? Look, Gant, remember that they're planning on keeping this thing running for a thousand years. They'll have to hide it good."

Gant shook his head with a wan smile. "Not at all. You forget that so far as anybody within the barytrine field is likely to see it, the total time will be from right now until the field goes on in a few hours. Then the enclosure-time will elapse instantaneously for those within. Anybody who finds it once the job goes on will find it after you have been freed of the field. The chances are high that they've dropped it in some comfortable climate, possibly near a large city, just as Scyth Radnor did."

Dusty eyed Gant sourly. "For the same purpose?" he asked.

"Probably. After all, Dusty—" Gant let the statement hang, suggesting silently that Dusty was the kind of human who would think of the same thing and act on it. "So you must issue orders to your Patrol—"

Dusty grunted. His Patrol? Discredited, his position shot to bits, his public appeal running somewhere near absolute zero, who would even listen to him? His former admirers had shucked their Space Patrol clothing for the costume of Jack Vandal, Space Rover.

Then he sat up with a puzzled smile.

"You have an idea?"

"I hope so."

"And—?"

Dusty smiled wistfully. "From the time Scyth Radnor opened his spacelock and blasted off the end of my antenna, I've been running a losing battle," he said. "I've been playing a game far over my head; outpointed, overbid,

overdrawn and sinking. About the only reason I'm still here fighting is that some of the rules of this cockeyed game seem to fall into my own act. Yes, dammit, I've got an idea. Can I call the orders, Gant?"

"Take over, Dusty."

Dusty turned to the pilot. "When we get there," he said, "Circle the planet several times as fast and as low as you can. Create a stir. Radiate like mad, anything you can radiate. Call attention to us in a bold fashion and show 'em that what we've got is big, important and powerful." Then to Gant Nerley he put the question, "You wouldn't have anything as primitive as a radio set aboard, would you?"

"You mean a radiomagnetic communication device? Well, not for communications but we do have a small receiver for detecting the lower-radiation stars and one for scanning planetary systems for primitive cultures. What did you have in mind?"

Dusty looked Gant in the eye. "I want to broadcast orders to my Patrol."

"Oh. An excellent idea. We'll save time that way. The scanner-type radiomagnetic wave equipment is two-way and connected to a menslator for contacting primitive peoples, you know, and—"

"Get it fired up," said Dusty shortly. "Full power."

The screech of air came first as a thin whistle, and then thundered and slammed down at Earth below as the thirteen Marandanian spacecraft were inched lower and lower into the complaining atmosphere. The howling racket dinned into the ears of Russian and Chinese and Hawaiian and Californian and New Yorker and Briton and Frenchman and Indian and Malayan and Indonesian and Argentinian and South African and Australian and Mexican and Floridian. Around it went, across the land and the sea, a thunder blast of rent air that piled shock wave on shock wave and sent them tearing down at the ground below. The thunder cracked windows and made plaster sift down from ceilings. It dinned down a tree or two, and it hurled some people to the ground. It flipped a parked fleet of jetplanes over in crumpled ruin like a windstorm hitting a deck of cards.

Across the world, radar operators looked blankly at the signal pips that raced across their screens and began to make apologetic reports. Interceptors tried to rise, but were tossed madly in the racing shock-stream to lose ground and return to earth limping.

But in the lead spacecraft of this mad fleet, the barytrine operator watched his detector hopefully. The entire screen was aglow, but he watched it and finally said, "I think it's down there somewhere."

He pointed to a region in Indiana not far from the lower tip of Lake Michigan.

The fleet circled Terra once more, swung high for the long dive, and then came howling down on a long slant, while Dusty took the radio and cried: "Junior Spacemen of The Space Patrol, *Attention!*"

The radio, powered by machinus forces, hammered down and blanketed the radio broadcast stations. It broke up the video screens in a mash of spots, flecks and snowflakes. Dusty's voice roared into telephone lines and onto the commercial radio links and chattered indistinctly in direction-finding equipment and made incomprehensible squiggles clutter the radar screens.

"Junior Spacemen, Attention to Official Orders! By now you are aware that your Commander, Dusty Britton, flies with a fleet of spacecraft above you. Now hear this!

"Within a few hundred miles of the lower tip of Lake Michigan there is concealed somewhere a dangerous device known as a barytrine generator. This must be located and stopped.

"Now! To the Junior Spaceman who locates this machine I will personally award the Medal of Merit. And to the entire Group Command of which he is a member I will award full scholarships as Space Midshipmen in a real Space Academy, to make them real spacemen.

"Now, Junior Spacemen, go out and find me that barytrine generator!"

Dusty signed off as the down-rushing fleet swaybacked close to the ground and pulled out to swap ends and go screaming up in a stark vertical climb, its drivers fighting the rise to a standstill fifty miles in the sky.

Here they hovered for a second to turn rightside up and then the flight formed into a pattern and began to land, coming down slowly.

Before they were halfway down, Dusty saw results. In the telescope were moving dots scouring the landscape. And along highways that led from town and city were boys on bicycles and a few in cars driven by parents. Across the fields they went, peering under trees and behind bushes, scouring the cornfields and the farms and stamping through woodsy sections like swarming ants.

But then as the flight landed in a neat pattern in a bald field, the barytrine detector hissed once and gave up, smoke curling out of the cabinet.

"Close," said the operator.

But Dusty, with a yell, was at the airlock. For across the field a thousand yards away was a faint bluish haze that shimmered iridescent in the sunlight. He pawed at the door as it swung open ponderously, then he looked around wildly for something to use. His eyes fell upon a small cabinet.

Scyth had placed that fluted-barrelled thing back in the airlock after he burned Dusty's antenna off. Dusty tore a cabinet open and grabbed one of the fluted-barrelled things from a clip.

Then he jumped to the ground and raced across the field.

"Dusty!" roared Gant Nerley. "That's dangerous. You can't—"

Gant let his voice trail away as Dusty plunged into the blue haze, fingering the trigger-button at the top of the pistol grip. The searing beam lashed out and slashed at the air as Dusty's heels caught the ground in a braking slide. Then the knifing beam slashed down across the metal case and into the ground before it. Curls of smoke arose and the ground sizzled. He cross-slashed and cut another ribbon out of the air and the barytrine generator, then cut down again.

There was a hiss and a sputter and the blue haze ceased—there was a blinding flash and a flat bark of something exploding violently. Dusty felt a wave of almost-intolerable heat, his closed eyes were seared by a flare of brightness, and the explosion hurled him backwards on his spine. He turned and scrambled back, stumbling over the rough ground, blinded.

At that moment four members of the Junior Space Patrol came through a small thicket of trees.

"Gee," said their Group Leader. "Gee—the Commander found it first!"

They stood on a small reviewing stand, Dusty Britton and the Group Command that had come through the thicket of trees in time to steer their blinded Commander away from the flaring barytrine generator. Dusty's face and hands were a super-sunburned red, and his eyes were still puffy but open enough to see.

From a sheet of paper he read:

"It is not within my power to grant a medal that is worth the tin it is made of. But for the diligence and their quick action I do hereby grant and guarantee them full scholarships in White Sands University, which by the time they graduate will have become a full Space Academy. So I here hand them their Certificates of Entry, and to the President of White Sands University I deliver a certified check to be held in trust and used for their education.

"I salute the future Commanders of The Space Patrol and step down from my position to leave it open for them!"

There came a roar from the crowd that thundered across the field as Dusty stepped from the platform into a spaceport jeep and was hustled out to Gant Nerley's flagship. Inside there were a number of men waiting.

"Now see here, Dusty, you can't go galaxy-hopping when we've got plans for you."

Dusty eyed Martin Gramer with a grunt. "Last time we met in a place like this you had me all scheduled to take a space hop when I had other plans for myself. No dice, Gramer."

"But look at the money—"

"I'll make millions out of this clear-channel idea, according to Gant, here."

"That's right," said Gant.

"So," said Dusty, "if you think I'm going to go on playing the part of a broken-down hero-spaceman when there are real spacemen around, you're

nuts, Gramer. Include me—as you've said so often—out."

"But what are you going to do?"

"Me? I'm going to Marandis. Barb and I have an offer from Supergalaxy Spectacles to make a series of what they call 'Primitives.' You know, old-timers with men using chemical rockets and learning their first feeble steps into space."

He grinned at Barbara knowingly. "I've got a script of *Destination Moon* I swiped from Central Files. It should oughta wow 'em cold!"

[Transcriber's Note: No Chapter XII heading in original publication.]

*** END OF THE PROJECT GUTENBERG EBOOK TROUBLED STAR

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